Ownership Structure and Economic Growth

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Abstract

This paper examines how the rise of common ownership affects economic growth and social welfare. We develop an endogenous growth model that incorporates three inter-firm networks: ownership, product-market rivalry, and technological spillovers. In the model, a large number of oligopolistic firms make forward-looking R&D investment decisions, internalizing the business stealing and technology spillovers that their R&D imposes on commonly owned firms. We quantify the model using data on 638 publicly traded U.S. firms with patents. Relative to an economy in which only R&D decisions reflect 2004 ownership intensity, the 2019 economy has 21.2 percent less R&D, a 0.087 percentage point lower growth rate, and 0.32 percent lower social welfare in consumption-equivalent terms, but a higher total value of firms. R&D is lower because the internalization of business stealing outweighs that of technology spillovers. Firm value is higher but welfare lower because common owners account for the profits that R&D diverts from their other firms but not for its gains to consumers and workers. Because firms already invest too little in R&D, the fall in R&D lowers welfare.

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1 Introduction

Over the past few decades, a small set of diversified institutional investors has come to hold sizable stakes in many competing U.S. firms at the same time, a trend often referred to as the rise of common ownership (Backus et al., 2021). At the end of 2017, BlackRock, Vanguard, and State Street together held on average about 20 percent of the shares of S&P 500 firms and cast about 25 percent of the votes at their annual meetings (Bebchuk and Hirst, 2019).[1] When common owners hold stakes in competing firms, managers seeking to maximize investor returns may partially internalize the externalities their actions impose on rivals' firm values (Rotemberg, 1984; Azar and Ribeiro, 2022). As a result, researchers and policymakers have become increasingly concerned about the implications of common ownership for product-market competition and firms' R&D activities (Ederer and Tecu, 2025). Guideline 11 of the 2023 Merger Guidelines (U.S. Department of Justice and Federal Trade Commission, 2023) notes that a partial owner may forgo developing a new product feature that would win market share from a firm in which it holds an interest.

A firm's R&D generates two inter-firm externalities (Bloom et al., 2013). It takes market share and profits from product-market rivals, a negative externality known as business stealing, and raises the productivity of firms that use similar technologies, a positive externality known as technology spillovers. Micro-level evidence shows that common ownership induces firms to partially internalize both forces (Antón et al., 2025; Kini et al., 2024). Ederer and Pellegrino (2026) analyze how common ownership affects the allocative efficiency of product markets through its impact on markups and competition, but we still lack a framework that quantifies how common ownership, via its effects on firms' R&D incentives, shapes aggregate productivity growth. In particular, it remains unclear whether the rise of common ownership leads firms to internalize positive or negative innovation externalities more strongly and, in turn, whether common ownership ultimately promotes or hinders economic growth and social welfare. We answer these questions with an endogenous growth model in which hundreds of oligopolistic firms, linked by measured ownership, product-market rivalry, and technological spillover networks, make forward-looking R&D decisions. The model allows the ownership weights that govern production and R&D decisions to differ, which separates the effect of common ownership through product-market conduct from its effect through R&D.

We find that the rise of common ownership has lowered growth through R&D decisions. Between 2004 and 2019, the average weight that a firm places on another firm's value

[1] Ownership links among large firms are not unique to the United States. For example, in South Korea the four largest conglomerates (the "chaebols") accounted for about 76 percent of the country's stock market capitalization in May 2026 (Seoul Economic Daily, 2026).

relative to its own rose from 0.20 to 0.42. Relative to an economy in which only R&D decisions reflect 2004 ownership intensity, the 2019 economy has 21.2 percent less R&D and a 0.087 percentage point lower growth rate, or 11 percent of the growth that R&D contributes in the observed economy. In the measured networks, the internalization of business stealing outweighs that of technology spillovers. Social welfare is 0.32 percent lower, while the total value of firms is 1.08 percent higher. The two move apart because common owners account for the profits that a firm's R&D diverts from rivals in which they also hold stakes but not for the gains that R&D brings to consumers and workers. The size of the effect depends on where ownership weights sit across firms. The largest R&D performers place the largest weights on other firms' values, and spreading the same average weight evenly across firms would shrink the growth effect by 68 percent.

Separating the ownership weights in production (product-market) and R&D decisions shows that the two margins move growth in opposite directions. Starting from the 2004 intensity in both decisions, raising ownership in production decisions softens competition and raises the growth rate by 0.036 percentage points, whereas raising it in R&D decisions lowers the growth rate by 0.079 percentage points. Softer competition, however, lowers the level of output, so production decisions account for 0.95 of the 1.27 percent welfare cost of common ownership. Whether common ownership changes firm conduct remains debated (Azar et al., 2018; Koch et al., 2021; Lewellen and Lowry, 2021; Dennis et al., 2022). Hau et al. (2026) find that price-cost margins rise with common ownership when firms' competitors are defined from the text-based product similarity of Hoberg and Phillips (2016), which we also use to measure product-market rivalry, but not in data aggregated to four-digit NAICS industries. Given this debate, we also consider models of corporate governance in which owners' influence on managers is weaker or stronger than proportional to their voting shares. The sign of the main result, in which ownership in production decisions stays at its 2019 level, is the same under alternative parameter values, ownership data, and models of corporate governance, and its size ranges from 0.029 to 0.130 percentage points across governance models.

In the model, firms accumulate knowledge capital through their own R&D investments, while technology spillovers from other firms augment the knowledge capital. The resulting knowledge capital raises product quality and labor productivity, thereby linking firms' R&D incentives to aggregate productivity growth. Business stealing runs through the product-market rivalry network and through the real wage, which a firm's expansion raises for all firms, whereas the technological spillover network governs the strength of technology spillovers. Owners receive profits in proportion to their shares, and each firm maximizes the value of its owners' portfolios, as in Rotemberg (1984) and Ederer and

Pellegrino (2026). Each firm thus maximizes a weighted sum of its own value and the values of other commonly owned firms, with weights given by the ownership network.

Our model extends the static product-market competition framework of Pellegrino (2025) and Ederer and Pellegrino (2026). Pellegrino (2025) applies a widely used hedonic demand system from the empirical IO literature in a general-equilibrium environment and develops an associated estimation strategy. Building on this static block, we follow Hopenhayn and Okumura (2026) in embedding firms' forward-looking R&D decisions and technology spillovers. Firms' R&D decisions then form a linear-quadratic differential game, in which firms' values are quadratic and their R&D policies linear in the vector of knowledge capital. The computational cost of the equilibrium therefore grows polynomially rather than exponentially in the number of firms, so we can solve the model with all 638 firms in the sample, each of which makes forward-looking R&D decisions that respond to all other firms. Because the ownership weights only reweight quadratic profit and cost terms, the game with common ownership remains linear-quadratic and keeps this tractability.

We discipline the model with data on 638 U.S. publicly traded patenting firms and measure the rise in common ownership from 2004 to 2019. Following Pellegrino (2025), we measure product-market rivalry between each pair of U.S. patenting firms with Hoberg and Phillips (2016)'s text-based product similarity. To construct the technological spillover network, we follow Bloom et al. (2013) and compute technological proximity from patent classification data. Corporate ownership is taken from the data of Amel-Zadeh et al. (2022), which combine the SEC filings of institutional investors, blockholders, and insiders. Combining these networks with Compustat financials, we also recover the model-implied distribution of firm-level knowledge capital. We take the strength of technology spillovers from Hopenhayn and Okumura (2026), who estimate it from firms' knowledge accumulation.

We also use the model to study R&D subsidies, comparing a uniform subsidy with the allocation of a planner who, like the subsidy, changes only R&D, as Hopenhayn and Okumura (2026) do in an economy without ownership links. Firms invest too little in R&D relative to this allocation, and the rise in common ownership widens this gap, raising the welfare-maximizing uniform subsidy from 31 to 37 percent. A regression of the ratio of social to private marginal returns to R&D on firms' centralities in the three networks shows that a uniform subsidy over-rewards firms that are central in the product-market rivalry network and under-rewards firms that are central sources of technology spillovers or that place large weight on other firms' values through ownership.

This study contributes to two literatures. First, we contribute to research on innovation and product-market competition (d'Aspremont and Jacquemin, 1988; Aghion et al., 2001,

2005; Acemoglu and Akcigit, 2012; Bloom et al., 2013; Peters, 2020; Akcigit and Ates, 2021, 2023; Liu et al., 2022; Cavenaile et al., 2026; De Ridder, 2024). Bloom et al. (2013) empirically document how product-market and technological proximity shape firms' R&D activity. In their estimates, higher product-market proximity strengthens business-stealing incentives, whereas greater technological proximity amplifies technology spillovers. Models with dynamic oligopoly, however, face a curse of dimensionality, with computational costs growing exponentially in the number of firms. For this reason, Cavenaile et al. (2026) develop a Schumpeterian growth model in which at most four firms compete within each industry, so strategic interaction arises only within industries. Because technology spillovers also flow across industries (Liu and Ma, 2024), their framework cannot analyze how common owners internalize these spillovers. To address this challenge, Hopenhayn and Okumura (2026) embed a linear-quadratic network game in an endogenous growth model with knowledge accumulation. Their model remains tractable with hundreds of firms and captures externalities both within and across industries. Relative to existing endogenous growth models, this approach directly links firm- and firm-pair-level data to the model. In concurrent work, Bustamante and Pellegrino (2026) extend the Q theory of investment to a network of product-market rivals, in which each firm's investment affects that of its rivals, and use it to study aggregate investment, cost-of-capital shocks, monopoly rents, and mergers. Complementing their analysis, we study R&D, which also generates technology spillovers, and our model adds an ownership network that governs production and R&D decisions separately. Ownership links between firms in different NAICS four-digit industries account for 92 percent of the growth effect in our calibration, and a framework with strategic interaction only within industries cannot capture them. Our analysis of R&D subsidies relates to estimates that the social return to R&D exceeds its private return (Bloom et al., 2013; Lucking et al., 2019) and to work on allocating R&D across an innovation network (Liu and Ma, 2024), and asks how common ownership changes the gap between the two returns across firms.

Second, our research contributes to the growing literature on common ownership and innovation (Gutiérrez and Philippon, 2017; He and Huang, 2017; Lopez and Vives, 2019; Antón et al., 2023, 2025; Kini et al., 2024). Lopez and Vives (2019) show in a static symmetric oligopoly that common ownership raises R&D when technology spillovers are large and lowers it when they are small. Antón et al. (2025) show that common ownership negatively affects innovation when firms are close in product markets but positively affects innovation when they are technologically proximate. They interpret this pattern as evidence of partial internalization of business stealing and technology spillovers. Using mergers among financial institutions as shocks to common ownership, Kini et al. (2024) find that common

ownership raises investment and new product development when technology spillovers are high. Our model contains both forces that these studies document, and aggregated over the measured networks, the internalization of business stealing dominates. Kasperk et al. (2024) report evidence that common ownership of product-market rivals lowers total factor productivity by reducing performance-sensitive managerial compensation, a channel distinct from the R&D incentives in our model. Unlike the static models of Lopez and Vives (2019), Azar and Vives (2021), and Ederer and Pellegrino (2026), we develop a dynamic general-equilibrium framework in which the growth rate is determined endogenously by firms' R&D decisions and ownership can govern production and R&D decisions separately.

The remainder of the paper proceeds as follows. Section 2 constructs an endogenous growth model incorporating common ownership. Section 3 describes the data and the calibration. Section 4 presents the quantitative results, including the analysis of R&D subsidies. Section 5 examines alternative corporate governance models. Section 6 concludes.

2 Model

We develop an endogenous economic growth model that incorporates three inter-firm networks: ownership, product-market rivalry, and technological spillovers. The demand system, the accumulation of knowledge capital, and the linear-quadratic R&D game follow Hopenhayn and Okumura (2026). This paper adds the ownership weights $\boldsymbol{K}^Q$ and $\boldsymbol{K}^R$, which govern production and R&D decisions.

2.1 Environment

There are n oligopolistic firms, indexed by $i \in \{1, 2, \ldots, n\}$, and they determine production and dynamic R&D investment under strategic interaction through the networks. Time is infinite and continuous, $t \in [0, \infty)$. The set of firms, the product-market rivalry and technological spillover networks, and the ownership weights are fixed over time. The model therefore abstracts from entry. Common ownership can also affect competition through entry, which becomes less likely when an incumbent and a potential entrant share owners (Newham et al., 2025).

2.1.1 Demand

We adopt the Generalized Hedonic Linear demand system of Pellegrino (2025) to model product-market rivalry among a large number of oligopolists. Products have two types

of characteristics. Common characteristics, indexed by $k \in \{1, 2, \ldots, m\}$, are shared by all products, and each product also has an idiosyncratic characteristic that shares the product's index i. The scalar $\psi_{ki} \geq 0$ denotes the quantity of the common characteristic k provided by product i. Each product i is described by the m-dimensional column vector $\boldsymbol{\psi}_i \equiv (\psi_{1i}, \ldots, \psi_{mi})^\top$ with nonnegative entries. The Euclidean length of the common-characteristics vector for each product is normalized to one without loss of generality,

$$\sum_{k=1}^{m} \psi_{ki}^2 = 1 \quad \forall i \in \{1, 2, \ldots, n\}.$$

Stacking these vectors across products gives the $m \times n$ matrix $\boldsymbol{\Psi} \equiv \begin{bmatrix} \boldsymbol{\psi}_1 & \cdots & \boldsymbol{\psi}_n \end{bmatrix}$, whose columns are products and whose rows are common characteristics. We aggregate the characteristics of all products to obtain the production function of the final-goods producer. Let $q_{i,t}$ denote the quantity of variety i produced by firm i and sold to the final-goods producer at time t, and let $\boldsymbol{q}_t \equiv (q_{1,t}, \ldots, q_{n,t})^\top$. Let $y_{k,t}^C$ be the total units of common characteristic k aggregated over all products,

$$y_{k,t}^C = \sum_{i=1}^{n} \psi_{ki} q_{i,t},$$

so that the matrix $\boldsymbol{\Psi}$ transforms units of goods into units of common characteristics,

$$\boldsymbol{y}_t^C = \boldsymbol{\Psi} \boldsymbol{q}_t. \tag{1}$$

Let $y_{i,t}^I$ be the number of units of idiosyncratic characteristic i, and assume that each unit of good i provides one unit of its corresponding idiosyncratic characteristic, so that

$$\boldsymbol{y}_t^I = \boldsymbol{q}_t. \tag{2}$$

A representative final-goods producer aggregates common and idiosyncratic characteristics using the linear-quadratic aggregator

$$Y_t = \alpha \sum_{k=1}^{m} \left(b_{k,t}^C y_{k,t}^C - \frac{1}{2} \left(y_{k,t}^C \right)^2 \right) + (1 - \alpha) \sum_{i=1}^{n} \left(b_{i,t}^I y_{i,t}^I - \frac{1}{2} \left(y_{i,t}^I \right)^2 \right), \tag{3}$$

where $b_{k,t}^C > 0$ and $b_{i,t}^I > 0$ are characteristic-specific quality, and $\alpha \in [0, 1]$ is the weight that is assigned to common characteristics. Inserting equations (1) and (2) into equation (3),

we obtain aggregate output in terms of the quantity vector,

$$Y_t = \boldsymbol{q}_t^\top \boldsymbol{b}_t - \frac{1}{2}\boldsymbol{q}_t^\top \boldsymbol{\Sigma} \boldsymbol{q}_t, \tag{4}$$

where

$$\boldsymbol{b}_t^C \equiv (b_{1,t}^C, \ldots, b_{m,t}^C)^\top, \quad \boldsymbol{b}_t^I \equiv (b_{1,t}^I, \ldots, b_{n,t}^I)^\top, \quad \boldsymbol{b}_t \equiv \alpha \boldsymbol{\Psi}^\top \boldsymbol{b}_t^C + (1-\alpha)\boldsymbol{b}_t^I,$$

$$\boldsymbol{\Sigma} \equiv \alpha \boldsymbol{\Psi}^\top \boldsymbol{\Psi} + (1-\alpha)\, \boldsymbol{I}, \tag{5}$$

and $\boldsymbol{I}$ is the $n \times n$ identity matrix. The remainder of the model works directly with the product-level quality vector $\boldsymbol{b}_t$. Thus, when firms choose $b_{i,t}$ below, the choice variable is the reduced-form quality that enters inverse demand, rather than the underlying characteristic-level qualities $\boldsymbol{b}_t^C$ and $\boldsymbol{b}_t^I$ separately.

$\boldsymbol{\Sigma}$ denotes the product-market rivalry matrix; each entry σ_{ij} measures substitutability between products i and j. As implied by equation (4), larger σ_{ij} increases the penalty in final-good production when both q_i and q_j are high. By construction, $\sigma_{ii} = 1$ for all i. The matrix $\boldsymbol{\Sigma}$ is a weighted sum of product-market proximity $\boldsymbol{\Psi}^\top \boldsymbol{\Psi}$ and the identity. Elements of $\boldsymbol{\Psi}^\top \boldsymbol{\Psi}$ are given by $[\boldsymbol{\Psi}^\top \boldsymbol{\Psi}]_{ij} = \boldsymbol{\psi}_i^\top \boldsymbol{\psi}_j$, i.e., the cosine similarity of the common characteristics of products i and j (since $\boldsymbol{\psi}_i^\top \boldsymbol{\psi}_j$ is the cosine of the angle between $\boldsymbol{\psi}_i$ and $\boldsymbol{\psi}_j$ in $\mathbb{R}^m$). From equation (5), for $i \neq j$ we have $\sigma_{ij} \propto \boldsymbol{\psi}_i^\top \boldsymbol{\psi}_j$, with the common-characteristics weight α scaling similarity into substitutability. As shown by Pellegrino (2025), this structure enables estimation of substitutability using product-similarity data.

2.1.2 Technology

Each firm has a production technology that is linear in labor,

$$q_{i,t} = a_{i,t} l_{i,t} \tag{6}$$

where $a_{i,t}$ is the labor productivity of firm i at time t and $l_{i,t}$ is the production labor employed by firm i at time t. As shown in equation (4), the quality $b_{i,t}$ is the value, measured in final good units, generated by the first unit of product i. Firms allocate their knowledge capital $z_{i,t}$ to improve labor productivity $a_{i,t}$ and product quality $b_{i,t}$ subject to

$$\zeta a_{i,t} + b_{i,t} = z_{i,t} \tag{7}$$

where $\zeta > 0$ is a parameter that determines the transformation rate between improvements in labor productivity and product quality[2].

We now introduce the law of motion for knowledge capital, which increases through technology spillovers from technologically similar firms and the firm's own R&D investment. Let $\widetilde{\boldsymbol{\Omega}} \equiv [\tilde{\omega}_{ij}]$ denote technological proximity, where $\tilde{\omega}_{ij} \geq 0$ is the weight of source firm j in firm i's spillovers, $\tilde{\omega}_{ii} = 0$, and each row sums to one (Section 3). Let $\boldsymbol{x}_t \equiv (x_{1,t}, \ldots, x_{n,t})^\top$ denote the vector of R&D efforts. Knowledge capital evolves according to

$$\dot{\boldsymbol{z}}_t = (\boldsymbol{\Omega} - \delta \boldsymbol{I})\, \boldsymbol{z}_t + \mu \boldsymbol{x}_t \tag{8}$$

where $\boldsymbol{\Omega} = \beta \widetilde{\boldsymbol{\Omega}}$ is the technological spillover network, $\beta \geq 0$ scales these weights into spillover intensities, $\mu \geq 0$ is the efficiency of R&D, common to all firms, and $\delta \geq 0$ is the rate of obsolescence of knowledge capital. Section A.8 extends the model to idiosyncratic shocks to knowledge capital.

2.1.3 Ownership Structure and Firm Objective Function

In our model, owners hold shares in multiple firms, and firms compete to maximize the weighted returns of their owners, with the weights determined by the owners' voting rights. There are n_o owners indexed by o. Let $\hat{V}^i$ denote the value of firm i, the discounted value of its cash flows. The portfolio value of owner o sums the values of the firms it holds, weighted by its shares of their cash flows,

$$\tilde{V}^o \equiv \sum_{i=1}^{n} s_{io} \hat{V}^i \tag{9}$$

where s_{io} is the proportion of firm i's cash-flow rights held by owner o such that $\sum_{o=1}^{n_o} s_{io} = 1$. Following Rotemberg (1984) and Ederer and Pellegrino (2026), we assume that firm i maximizes a weighted sum of its owners' portfolio values. Because owners influence a firm's decisions through their votes, the weights are their voting shares in firm i, and the objective is $\sum_{o=1}^{n_o} v_{io} \tilde{V}^o$, where v_{io} is the proportion of firm i's voting rights held by owner o such that $\sum_{o=1}^{n_o} v_{io} = 1$. Substituting the portfolio value (9) expresses this objective as a

[2]Assuming optimal allocation of knowledge capital between labor-productivity and quality improvements is necessary for a balanced-growth-path equilibrium. Similarly, Grossman et al. (2017) and Jones and Liu (2024) obtain a balanced growth path by assuming two productivity-enhancing technologies and allocating resources between them (capital and human capital; capital productivity and automation).

linear combination of firm values $\hat{V}^j$,

$$\sum_{o=1}^{n_o} v_{io}\tilde{V}^o = \sum_{j=1}^{n} \hat{V}^j \, \boldsymbol{v}_i^\top \boldsymbol{s}_j = \left(\boldsymbol{v}_i^\top \boldsymbol{s}_i\right) \sum_{j=1}^{n} \kappa_{ij}\hat{V}^j,$$

where

$$\kappa_{ij} \equiv \frac{\boldsymbol{v}_i^\top \boldsymbol{s}_j}{\boldsymbol{v}_i^\top \boldsymbol{s}_i} \tag{10}$$

and $\boldsymbol{v}_i \equiv (v_{i1}, \ldots, v_{i,n_o})^\top$ and $\boldsymbol{s}_i \equiv (s_{i1}, \ldots, s_{i,n_o})^\top$. Voting and cash-flow rights differ for firms with multiple share classes. When they coincide, κ_{ij} reduces to $\boldsymbol{s}_i^\top \boldsymbol{s}_j / \boldsymbol{s}_i^\top \boldsymbol{s}_i$ of Rotemberg (1984). Because $\boldsymbol{v}_i^\top \boldsymbol{s}_i > 0$ is a constant for firm i, this scale factor does not affect the maximization problem, and firm i equivalently maximizes

$$V^i \equiv \sum_{j=1}^{n} \kappa_{ij}\hat{V}^j.$$

We define the ownership weight matrix $\boldsymbol{K} \equiv [\kappa_{ij}]$. By construction, $\kappa_{ii} = 1$ for all i. Ownership weight κ_{ij} is the relative weight on firm j's value in firm i's objective. The objective function nests standard profit maximization ($\kappa_{ij} = 0$ for $i \neq j$) and monopoly ($\kappa_{ij} = 1$ for $i \neq j$) as limit cases.

We allow the ownership weights that govern production decisions to differ from those that govern R&D decisions. Let $\boldsymbol{K}^Q$, with entries κ_{ij}^Q, denote the ownership weights in firms' static production problem, and let $\boldsymbol{K}^R$, with entries κ_{ij}^R, denote the ownership weights in their dynamic R&D problem. We refer to them as production ownership and R&D ownership. Both matrices have unit diagonals. In the observed economy, both equal the measured matrix, $\boldsymbol{K}^Q = \boldsymbol{K}^R = \boldsymbol{K}$. Separating the two lets the counterfactuals in Section 2.4 change ownership in one decision while holding the other fixed. The distinction is also natural when R&D reflects the preferences of higher corporate layers, such as boards, whereas product-market quantities reflect those of lower layers, such as product managers.

2.1.4 Resources and Household Preferences

Production labor is supplied inelastically in a fixed amount $L > 0$,

$$L = \sum_i l_{i,t} \tag{11}$$

Final goods are allocated to consumption and R&D expenditures. Following the endogenous growth literature (e.g., Acemoglu et al. 2016), we assume an innovation elasticity of 1/2 at the firm level; with units normalized so that the cost coefficient equals one, total R&D expenditure is quadratic in effort, and the resource constraint is

$$C_t + \sum_i x_{i,t}^2 = Y_t \tag{12}$$

where C_t denotes final-good consumption at time t. The representative household ultimately owns all shares, is risk-neutral, and discounts the future at rate $\rho > 0$; lifetime utility is

$$\int_0^\infty \exp\left(-\rho t\right) C_t \, dt \tag{13}$$

2.2 Equilibrium

We focus on a Markov perfect equilibrium with knowledge capital z as the state. Because quantity and technology choices do not affect the evolution of knowledge capital, the problem separates into a static product-market problem and a dynamic R&D problem. We describe the final-goods producer and the two problems of firms in turn and then define the equilibrium.

2.2.1 Final-Goods Producer

A competitive representative final-goods producer chooses quantities at each instant to solve

$$\max_{\boldsymbol{q}_t} \; P_t Y_t - \boldsymbol{q}_t^\top \boldsymbol{p}_t$$

where $\boldsymbol{p}_t$ is the vector of product prices at time t and P_t is the final-good price. Profit maximization yields the linear inverse demand function

$$\frac{\boldsymbol{p}_t}{P_t} = \boldsymbol{b}_t - \boldsymbol{\Sigma} \boldsymbol{q}_t \tag{14}$$

2.2.2 Static Problem of Firms: Production Decision and Technology Choice

We assume that firms maximize the real returns of their owners to ensure independence of the allocation from the choice of numeraire (Azar and Vives, 2021). Each firm's real gross

profit *before* subtracting R&D cost, $\pi_{i,t}$, is given by

$$\begin{aligned}\pi_{i,t} &= \frac{p_{i,t}}{P_t} q_{i,t} - \frac{w_t}{P_t} l_{i,t} \\ &= \left(b_{i,t} - \frac{1}{a_{i,t}} \frac{w_t}{P_t} - \sum_j \sigma_{ij} q_{j,t} \right) q_{i,t} \end{aligned} \tag{15}$$

In the first line on the right-hand side, the real labor cost is subtracted from real revenue. The second line is obtained by substituting the inverse demand in equation (14) and the production function in equation (6).

In the static problem, the objective of firm i is the sum of real gross profits weighted by the production ownership weights,

$$\sum_j \kappa_{ij}^{Q} \pi_{j,t}$$

Each firm i simultaneously chooses quantity $q_{i,t}$ and the allocation of knowledge capital between labor productivity $a_{i,t}$ and product quality $b_{i,t}$, taking as given the output of other firms $\{q_{j,t}\}_{j \neq i}$, the real wage w_t/P_t, and its own knowledge capital $z_{i,t}$, subject to the knowledge-capital constraint (7). The first-order condition with respect to $q_{i,t}$ gives

$$0 = b_{i,t} - \frac{1}{a_{i,t}} \frac{w_t}{P_t} - 2q_{i,t} - \sum_{j \neq i} \sigma_{ij} q_{j,t} - \sum_{j \neq i} \kappa_{ij}^{Q} \sigma_{ji} q_{j,t} \tag{16}$$

From the optimality conditions for $a_{i,t}$ and $b_{i,t}$, we obtain

$$a_{i,t} = \sqrt{\frac{w_t}{\zeta P_t}} \tag{17}$$

$$b_{i,t} = z_{i,t} - \sqrt{\frac{\zeta w_t}{P_t}}. \tag{18}$$

Intuitively, firms invest more in labor productivity when labor costs w_t/P_t are higher. Inserting equation (17) into the labor-market clearing condition (11) yields

$$\sqrt{\zeta \frac{w_t}{P_t}} = \frac{\zeta}{L} \sum_i q_{i,t}. \tag{19}$$

Combining equations (16) to (19) yields a linear relationship between quantities and

knowledge capital,

$$\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t \tag{20}$$

where

$$\boldsymbol{N} \equiv \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{K}^Q \circ \boldsymbol{\Sigma}\right)^{-1},$$

$\boldsymbol{J}$ is an $n \times n$ matrix with all elements equal to one (see Section A.1 for the derivation). The first term inside the parentheses, $2\frac{\zeta}{L}\boldsymbol{J}$, captures labor costs; the second term $\boldsymbol{\Sigma}$ represents product-market rivalry; and the third term $\boldsymbol{K}^Q \circ \boldsymbol{\Sigma}$ is the element-wise product of production ownership weights and rivalry. Hence, in the static problem, ownership influences production only through its overlap with product-market rivalry.

Given the linear relationship between the quantity $\boldsymbol{q}_t$ and knowledge capital $\boldsymbol{z}_t$, the profit flow that enters firm i's R&D objective below, weighted by the R&D ownership weights, is a quadratic form of knowledge capital $\boldsymbol{z}_t$,

$$\sum_j \kappa_{ij}^R \pi_{j,t} = \boldsymbol{z}_t^\top \boldsymbol{Q}^i \boldsymbol{z}_t$$

where

$$\boldsymbol{Q}^i \equiv \sum_h \kappa_{ih}^R \sum_j \kappa_{hj}^Q \sigma_{hj} \frac{\boldsymbol{N}_h^\top \boldsymbol{N}_j + \boldsymbol{N}_j^\top \boldsymbol{N}_h}{2}.$$

and $\boldsymbol{N}_i$ denotes the i-th row of $\boldsymbol{N}$ (see Section A.2 for the derivation). The outer weights κ_{ih}^R determine whose profits firm i values when choosing R&D, whereas the inner weights κ_{hj}^Q determine each firm's profit through product-market conduct. In a symmetric economy, the two sets of weights move the operating-profit return to a firm's own knowledge capital in opposite directions. Stronger production ownership weakly raises this return because it softens product-market competition. Stronger R&D ownership lowers it because the firm internalizes the profit its knowledge takes from its rivals (Proposition 2 in Section A.4).

2.2.3 Dynamic Problem of Firms: R&D Decision

Taking the static production equilibrium as given, each firm i chooses a Markov R&D policy $x_i(\boldsymbol{z})$, given the other firms' policies $\left\{x_j(\boldsymbol{z})\right\}_{j \neq i}$, to maximize the discounted sum of

profits weighted by the R&D ownership weights,

$$\max_{x_i(\cdot)} \quad V^i(z_0) \equiv \int_0^\infty \exp(-\rho t) \left\{ \sum_j \kappa_{ij}^R \left(\pi_{j,t} - x_{j,t}^2 \right) \right\} dt$$

where $x_{j,t} = x_j(z_t)$ for every firm j and the state evolves according to equation (8). Firms discount at the interest rate, which equals ρ by the household's Euler equation. The HJB equation for firm i is

$$\rho V^i(z) = \max_{x_i} \left\{ z^\top \boldsymbol{Q}^i z - \sum_j \kappa_{ij}^R x_j^2 + V_z^i(z) \left[(\boldsymbol{\Omega} - \delta \boldsymbol{I}) z + \mu \boldsymbol{x} \right] \right\}. \tag{21}$$

The first two terms on the right-hand side are the ownership-weighted gross profit net of ownership-weighted R&D costs, and the third term reflects the effect of changes in knowledge capital due to technology spillovers and R&D on firm i's objective. Firm i chooses only x_i; the other entries of $\boldsymbol{x}$, including the rivals' R&D costs that firm i weights by κ_{ij}^R, are evaluated at their equilibrium policies.

Definition 1. Equilibrium: Given production ownership $\boldsymbol{K}^Q$ and R&D ownership $\boldsymbol{K}^R$, an *equilibrium* consists of Markov R&D policies, firm value functions, static allocations, product prices, a real wage, an interest rate, and consumption such that (i) the final-goods producer maximizes profits; (ii) each firm's quantity and technology choices maximize its static objective weighted by $\boldsymbol{K}^Q$, given the other firms' choices; (iii) each firm's R&D policy maximizes its dynamic objective weighted by $\boldsymbol{K}^R$, given the other firms' policies; (iv) knowledge capital evolves according to equation (8); (v) the labor market clears; (vi) the household maximizes lifetime utility (13), so the interest rate equals ρ; and (vii) the goods market clears according to the resource constraint (12).

We focus on linear Markov perfect equilibria, in which R&D policies are linear and value functions are quadratic in knowledge capital, with $\boldsymbol{X}^i$ symmetric,

$$V^i(z) = z^\top \boldsymbol{X}^i z, \tag{22}$$

with derivative

$$V_z^i(z) = 2 z^\top \boldsymbol{X}^i \tag{23}$$

Let $\boldsymbol{X}_i^i$ denote the i-th column of $\boldsymbol{X}^i$. The first-order condition with respect to x_i gives

$$x_i = \left(\mu \boldsymbol{X}_i^i\right)^\top \boldsymbol{z}. \tag{24}$$

Given firms' linear strategies, the law of motion for knowledge capital $\boldsymbol{z}_t$ becomes a linear system with constant coefficients. Let $\widetilde{\boldsymbol{X}}$ stack the policy columns $\boldsymbol{X}_i^i$,

$$\widetilde{\boldsymbol{X}} \equiv \begin{bmatrix} \boldsymbol{X}_1^1 & \cdots & \boldsymbol{X}_n^n \end{bmatrix}^\top \tag{25}$$

Then, the law of motion is rewritten as

$$\dot{\boldsymbol{z}}_t = \boldsymbol{\Phi} \boldsymbol{z}_t \tag{26}$$

where

$$\boldsymbol{\Phi} \equiv \boldsymbol{\Omega} + \mu^2 \widetilde{\boldsymbol{X}} - \delta \boldsymbol{I},$$

with entries ϕ_{ij}. The *technology transition matrix* $\boldsymbol{\Phi}$ combines the exogenous technological spillover network $\boldsymbol{\Omega}$, obsolescence at rate δ, and the endogenous R&D policies $\mu^2 \widetilde{\boldsymbol{X}}$.

Inserting equations (22) to (24) into the HJB equation (21) and rearranging (see Section A.5), we obtain the stacked algebraic Riccati equations

$$0 = \boldsymbol{Q}^i - \mu^2 \sum_j \kappa_{ij}^R \boldsymbol{X}_j^j \left(\boldsymbol{X}_j^j\right)^\top + \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right)^\top \boldsymbol{X}^i + \boldsymbol{X}^i \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right) \tag{27}$$

for $i = 1, 2, \ldots, n$. Because the R&D cost term and $\boldsymbol{\Phi}$ depend on the other firms' $\boldsymbol{X}^j$, the n Riccati equations must be solved jointly.

Definition 2. The solution of the stacked algebraic Riccati equations (27) is *stable* if all eigenvalues of $\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}$ have strictly negative real parts.

Stability requires every eigenvalue of $\boldsymbol{\Phi}$ to have a real part below $\rho/2$. Because payoffs are quadratic in knowledge capital, this condition keeps firm values and lifetime utility finite.

Assumption 1. *There exists a stable solution* $\left(\boldsymbol{X}^1, \boldsymbol{X}^2, \ldots, \boldsymbol{X}^n\right)$ *of the stacked algebraic Riccati equations* (27)*.*

When the stacked equations have more than one stable solution, we select the limit of finite-horizon equilibria as the horizon grows (Cartigny and Michel, 2003), which the backward iteration in Section B.1 computes.

2.2.4 Balanced Growth Path

The following assumption on the technology transition matrix $\mathbf{\Phi}$ is sufficient for a balanced growth path.

Assumption 2. *$\mathbf{\Phi}$ is irreducible, and its off-diagonal elements are nonnegative.*

The matrix $\mathbf{\Phi}$ is irreducible if the directed graph with an edge from firm j to firm i whenever $\phi_{ij} > 0$ is strongly connected, so that each firm's knowledge capital eventually affects that of every other firm, directly or indirectly. Nonnegative off-diagonal elements ensure that knowledge capital that starts nonnegative stays nonnegative.

Definition 3. Balanced Growth Path Equilibrium: A *balanced-growth-path equilibrium* is a linear Markov perfect equilibrium along which every firm's knowledge capital grows at a common constant rate.

Proposition 1. *Let Assumptions 1 and 2 hold. Then $\mathbf{\Phi}$ has a real dominant eigenvalue, denoted g, and a strictly positive eigenvector $\bar{\mathbf{z}}$ associated with g. Moreover, for any strictly positive initial knowledge capital, the normalized distribution of knowledge capital converges to a scalar multiple of $\bar{\mathbf{z}}$, and the knowledge capital of all firms grows asymptotically at rate g.*

Because $\mathbf{\Phi}$ contains the endogenous R&D policy $\mu^2\widetilde{\boldsymbol{X}}$, the long-run growth rate depends on firms' R&D decisions, and the associated eigenvector gives the long-run distribution of knowledge capital, unique up to scale. Along the balanced growth path, knowledge capital grows at rate g and output at rate $2g$, which is below ρ by Assumption 1. Section A.6 proves the proposition, reports the growth rates of all variables (Table A.1), and illustrates the path with a partial-equilibrium diagram.

In the observed economy, firms' R&D responds negatively to rivals' knowledge capital, so 69 percent of the off-diagonal elements of $\mathbf{\Phi}$ are negative and Assumption 2 does not hold. The quantitative analysis therefore evaluates the equilibrium at the observed knowledge capital and reports the current growth rate of output at that state (Section 4.2).

2.3 Welfare and Firm Value

We evaluate each equilibrium by the lifetime utility of the representative household and by the total value of firms.

2.3.1 Social Welfare

Let $V(z_0)$ denote the lifetime utility of the representative household in equilibrium, given initial knowledge capital z_0,

$$V(z_0) = \int_0^\infty \exp(-\rho t) \left\{ Y_t - \sum_i x_{i,t}^2 \right\} dt.$$

Total output is

$$Y_t = z_t^\top \boldsymbol{Q} z_t \tag{28}$$

where

$$\boldsymbol{Q} = \frac{1}{2} \boldsymbol{N}^\top \left(2\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{\Sigma} \circ \left(\boldsymbol{K}^{\mathcal{Q}} + \left(\boldsymbol{K}^{\mathcal{Q}} \right)^\top \right) \right) \boldsymbol{N}$$

(see Section A.3 for the derivation). Production ownership affects this quadratic form through the equilibrium allocation of quantities and qualities; R&D ownership affects welfare through the path of knowledge capital. Since equilibrium R&D is $\boldsymbol{x} = \mu \widetilde{\boldsymbol{X}} \boldsymbol{z}$, total R&D cost is $\boldsymbol{z}^\top \mu^2 \widetilde{\boldsymbol{X}}^\top \widetilde{\boldsymbol{X}} \boldsymbol{z}$. The representative household's HJB equation is

$$\rho V(z) = z^\top \left(\boldsymbol{Q} - \mu^2 \widetilde{\boldsymbol{X}}^\top \widetilde{\boldsymbol{X}} \right) z + V_z(z) \boldsymbol{\Phi} \boldsymbol{z}$$

The first term on the right-hand side of the HJB equation represents the household's instantaneous consumption, equal to output net of total R&D cost. The second term represents the change in the household's value due to the accumulation of knowledge capital.

Because the flow payoff is quadratic and the law of motion is linear, the value function is quadratic in knowledge capital z,

$$V(z) = z^\top \boldsymbol{X} z.$$

Substituting $V(z)$ and its derivative gives the algebraic Lyapunov equation (see Section A.5)

$$0 = \boldsymbol{Q} - \mu^2 \widetilde{\boldsymbol{X}}^\top \widetilde{\boldsymbol{X}} + \boldsymbol{X} \left(\boldsymbol{\Phi} - \frac{\rho}{2} \boldsymbol{I} \right) + \left(\boldsymbol{\Phi} - \frac{\rho}{2} \boldsymbol{I} \right)^\top \boldsymbol{X} \tag{29}$$

Therefore, $\boldsymbol{X}$ is obtained as the solution to the Lyapunov equation (29), and equilibrium welfare given initial knowledge capital z_0 is $V(z_0) = z_0^\top \boldsymbol{X} z_0$.

2.3.2 Total Value of Firms

Changes in ownership structure affect not only social welfare but also how it is divided between firms' owners on the one hand and workers and consumers on the other. To quantify this division, we characterize the total value of firms. Let V^P denote the total value of firms in equilibrium, defined as

$$V^P(\boldsymbol{z}_0) = \int_0^\infty \exp(-\rho t) \sum_i \left(\pi_{i,t} - x_{i,t}^2\right) dt.$$

Aggregate real profits can be written in vector form as

$$\begin{aligned}\sum_i \pi_{i,t} &= \sum_i \sum_j \kappa_{ij}^Q \sigma_{ij} q_{i,t} q_{j,t} \\ &= \boldsymbol{q}_t^\top \left(\boldsymbol{K}^Q \circ \boldsymbol{\Sigma}\right) \boldsymbol{q}_t \\ &= \boldsymbol{z}_t^\top \boldsymbol{\Pi} \boldsymbol{z}_t\end{aligned}$$

where

$$\boldsymbol{\Pi} = \frac{1}{2} \boldsymbol{N}^\top \left(\boldsymbol{\Sigma} \circ \left(\boldsymbol{K}^Q + \left(\boldsymbol{K}^Q\right)^\top\right)\right) \boldsymbol{N}.$$

Output exceeds aggregate profits by labor income, $\frac{\zeta}{L} \boldsymbol{q}_t^\top \boldsymbol{J} \boldsymbol{q}_t$, and the surplus of the final-goods producer, $\frac{1}{2} \boldsymbol{q}_t^\top \boldsymbol{\Sigma} \boldsymbol{q}_t$, because $\boldsymbol{Q} - \boldsymbol{\Pi} = \frac{1}{2} \boldsymbol{N}^\top \left(2\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma}\right) \boldsymbol{N}$. No owner's objective contains these terms. By analogy with the social-welfare calculation in Section 2.3.1, V^P is obtained by solving the Lyapunov equation (29) with $\boldsymbol{Q}$ replaced by $\boldsymbol{\Pi}$. Denote the solution by $\boldsymbol{X}^P$. Then the total value of firms is

$$V^P(\boldsymbol{z}_0) = \boldsymbol{z}_0^\top \boldsymbol{X}^P \boldsymbol{z}_0.$$

We report the percent change in V^P and its level as a share of household lifetime utility V, the firm value share.

2.4 Counterfactual Scenarios

The counterfactuals hold initial knowledge capital, the product-market rivalry and technological spillover networks, and all parameters fixed. They change either the ownership weights in firms' decisions or the mechanism that allocates R&D. We first define the ownership counterfactuals and then the R&D-only planner, a benchmark for R&D policy.

2.4.1 Ownership Counterfactuals

Let $\boldsymbol{K}_{2004}$ denote the measured ownership matrix with every off-diagonal weight scaled by the ratio of the 2004 mean off-diagonal weight to the mean off-diagonal weight of $\boldsymbol{K}$ (Section 3). This matrix preserves the observed placement of ownership links and lowers only their average intensity to its 2004 level.

Our main counterfactual changes ownership only in R&D decisions. It compares the observed economy, $\left(\boldsymbol{K}^Q, \boldsymbol{K}^R\right) = (\boldsymbol{K}, \boldsymbol{K})$, with the economy $(\boldsymbol{K}, \boldsymbol{K}_{2004})$, in which R&D ownership is at its 2004 intensity. Because production ownership is the same in the two economies, the comparison isolates how the rise in common ownership changes the externalities that firms internalize when they choose R&D, namely business stealing from product-market rivals and technology spillovers to technologically proximate firms.

To separate the two decision margins, we also solve the four combinations $\left(\boldsymbol{K}^Q, \boldsymbol{K}^R\right) \in \{\boldsymbol{K}_{2004}, \boldsymbol{K}\}^2$. Raising production ownership from $\boldsymbol{K}_{2004}$ to $\boldsymbol{K}$ softens product-market competition, which changes the private return to R&D. The four combinations split the difference between the observed economy and $(\boldsymbol{K}_{2004}, \boldsymbol{K}_{2004})$ into the effect of each margin, measured from $(\boldsymbol{K}_{2004}, \boldsymbol{K}_{2004})$, and their interaction (Section 4.5).

To distinguish the placement of ownership links from their average intensity, we also raise R&D ownership from $\boldsymbol{K}_{2004}$ to a uniform matrix $\bar{\boldsymbol{K}}$ instead of $\boldsymbol{K}$, keeping $\boldsymbol{K}^Q = \boldsymbol{K}$. The off-diagonal entries of $\bar{\boldsymbol{K}}$ all equal the mean off-diagonal entry of $\boldsymbol{K}$. As a benchmark without common ownership, we also solve fully dispersed ownership, $\boldsymbol{K}^Q = \boldsymbol{K}^R = \boldsymbol{I}$, in which each firm maximizes its own value.

2.4.2 R&D-Only Planner

To benchmark R&D policy, we consider an R&D-only planner, who takes as given the static equilibrium under production ownership $\boldsymbol{K}^Q$ and chooses only the R&D allocation. This is the relevant benchmark for R&D subsidies, which change R&D incentives but not product-market conduct. Starting from initial knowledge capital $\boldsymbol{z}_0$, the planner solves

$$V^{**}(\boldsymbol{z}_0) \equiv \max_{\{\boldsymbol{x}_t\}_{t\geq 0}} \int_0^\infty \exp\left(-\rho t\right) \left\{\boldsymbol{z}_t^\top \boldsymbol{Q} \boldsymbol{z}_t - \boldsymbol{x}_t^\top \boldsymbol{x}_t\right\} dt,$$

subject to the law of motion (8). The HJB equation of the planner's value V^{**} is

$$\rho V^{**}(\boldsymbol{z}) = \max_{\boldsymbol{x}} \left\{\boldsymbol{z}^\top \boldsymbol{Q} \boldsymbol{z} - \boldsymbol{x}^\top \boldsymbol{x} + V_{\boldsymbol{z}}^{**}(\boldsymbol{z}) \left[(\boldsymbol{\Omega} - \delta \boldsymbol{I})\, \boldsymbol{z} + \mu \boldsymbol{x}\right]\right\}. \tag{30}$$

Because the planner's problem is also linear-quadratic, the value is quadratic in

knowledge capital, $V^{**}(z) = z^\top X^{**} z$. The planner's R&D choice is not restricted in sign, so the planner's R&D is a linear function of knowledge capital, $x = \mu X^{**} z$, as in the market equilibrium. The planner's welfare gain is therefore an upper bound on the gain attainable with nonnegative R&D. Knowledge capital then follows $\dot{z}_t = \Phi^{**} z_t$ with

$$\Phi^{**} \equiv (\Omega - \delta I) + \mu^2 X^{**}.$$

Substituting the value function and the R&D policy into equation (30) gives a single algebraic Riccati equation,

$$0 = Q - \mu^2 (X^{**})^2 + X^{**} \left(\Phi^{**} - \frac{\rho}{2} I\right) + \left(\Phi^{**} - \frac{\rho}{2} I\right)^\top X^{**} \tag{31}$$

Social welfare under the planner's allocation is $V^{**}(z_0) = z_0^\top X^{**} z_0$. Section A.7 describes the social planner who also chooses production.

3 Data and Calibration

This section maps the model's parameters, including the product-market rivalry, technological spillover, and ownership networks, to data. The product-market rivalry and technological spillover networks and the recovery of knowledge capital follow Hopenhayn and Okumura (2026), and we add the ownership network. Table 1 summarizes the parameters and their sources.

3.1 Sample Construction

The quantitative model is calibrated to U.S. publicly traded firms in 2019. We begin with firms that can be linked across Compustat, DISCERN 2.0, the Hoberg–Phillips product-similarity data, and the ownership data described in Section 3.2. This restriction ensures that, for each sample firm, we observe accounting variables, patent-based technology classes, product-market proximity, and ownership.

We then impose two additional restrictions. First, we retain firms with positive gross profits, measured as revenue minus cost of goods sold, because the model's static profit identity is used to recover positive quantities. Second, we require at least ten patents in the year or within two years on either side. This five-year window reduces noise in patent-class shares. The resulting sample contains 638 firms. The sample firms account for 22 percent of the sales and 62 percent of the R&D expenditure of all publicly traded Compustat firms with positive sales in 2019. To measure how common ownership has risen, we construct the

Table 1: Model Parameters

Notation	Description	Value	Source
$\boldsymbol{K}$	Ownership weights		SEC ownership filings, Amel-Zadeh et al. (2022)
$\boldsymbol{\Psi}^\top\boldsymbol{\Psi}$	Product-market proximity		Form 10-K, Hoberg and Phillips (2016)
α	Product-market proximity → Substitutability	0.12	Pellegrino (2025)
$\widetilde{\boldsymbol{\Omega}}$	Technological proximity		DISCERN 2.0 patents, CPC main groups, Arora et al. (2024)
β	Technological proximity → Spillover	0.02	Hopenhayn and Okumura (2026)
ζ/L	Labor-augmentation efficiency	0.0074	Compustat, cost of goods sold
ρ	Discount rate	0.10	Firms' reported discount rates, Poterba and Summers (1995), Gormsen and Huber (2025)
μ	R&D efficiency	0.0699	R&D share of business value added (joint moment match)
δ	Obsolescence rate	0.0136	Growth of real GDP per capita (joint moment match)

sample for each year from 2004 to 2019 in the same way and compute ownership weights among the firms in each year's sample.

3.2 Ownership Weights

To calculate the matrix of ownership weights $\boldsymbol{K}$, we use the ownership data of Amel-Zadeh et al. (2022). The data combine three types of filings with the Securities and Exchange Commission (SEC): Form 13F, which institutional investors with over $100 million in assets under management file to disclose their holdings of U.S. securities; Schedules 13D and 13G, which any owner of more than five percent of a firm's shares files; and Forms 3, 4, and 5, which officers, directors, and owners of more than ten percent of a firm's shares file. The data therefore cover blockholders and insiders in addition to institutional investors. For each firm and owner, the data report the owner's share of the firm's cash-flow rights, s_{io}, and of its voting rights, v_{io}. We use holdings as of June 30 of each year. We drop 13F positions from filings made before June 30, which the data carry forward from earlier quarters even after the institution has stopped reporting them. For 2004 and 2019, the two years that the model uses, we also correct holdings, mostly of insiders and blockholders, that differ from proxy statements and other SEC filings.[3]

For 24 percent of sample firms, the observed cash-flow or voting shares sum to more than one, with a median total of 106 percent among these firms. Totals above one arise

[3]The replication package lists each correction with the filing it rests on.

when the same shares are counted in more than one filing or when holdings and shares outstanding are measured inconsistently. In these firms, 13F filings alone already account for 93 percent of shares on average. We scale the shares proportionally so that they sum to one.

Observed owners hold 92 percent of the median sample firm's shares. Because blockholders and insiders are observed, the remaining owners are mainly small, dispersed shareholders, whose holdings contribute little to the denominator $\boldsymbol{v}_i^\top \boldsymbol{s}_i$. We therefore compute κ_{ij} from the observed holdings without adjusting for unobserved owners.

As shown in Figure 1, the average ownership weight κ_{ij} rose from 0.20 in 2004 to 0.42 in 2019. Within our framework, this implies that firms increasingly took into account the values of other firms when determining production and R&D activity during this period. The rise comes from the growth of the three largest index fund managers, BlackRock, Vanguard, and State Street.[4] Because the numerator of κ_{ij} is a sum over owners, the average weight splits into the parts due to each owner's holdings. The part due to the three managers rose from 0.08 to 0.34, or 81 percent of the 2019 average, while the part due to all other owners fell from 0.12 to 0.08. Behind this rise, the three managers' combined stake in the average sample firm grew from 7 percent in 2004 to 19 percent in 2019.

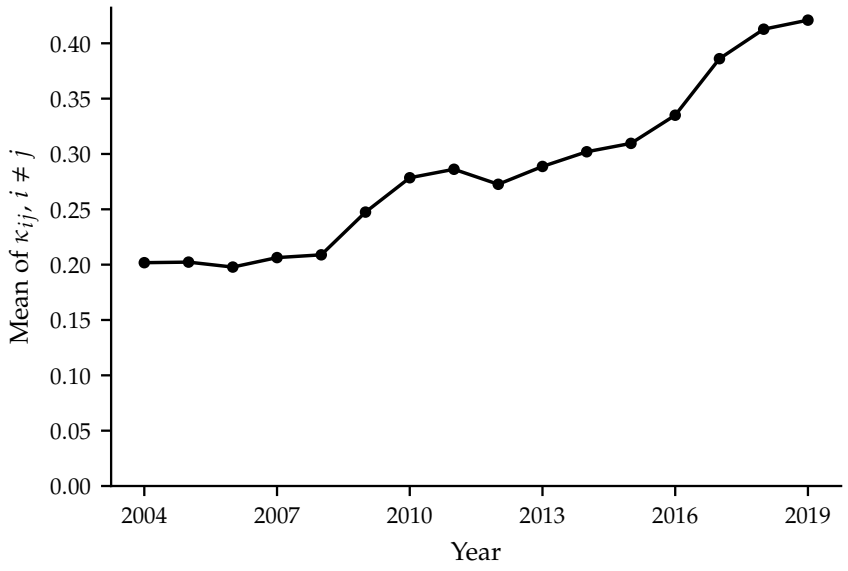
The ratio of the two averages, 0.48, is the factor that scales the observed matrix to $\boldsymbol{K}_{2004}$ (Section 2.4.1). We scale the observed matrix rather than use the matrix measured in 2004 because the two years differ in both their firms and the placement of their ownership links. Only 287 of the 638 firms in the 2019 sample are also in the 2004 sample, and among these firms the correlation of κ_{ij} between the two years is 0.43. Among the firms in both samples, which account for 77 percent of the 2019 sample's sales and 71 percent of its R&D expenditure, the ratio of mean weights is 0.42, somewhat below the full-sample ratio of 0.48, so the rise in average ownership does not reflect changes in the composition of the sample.

3.3 Product-Market Proximity and Substitutability

Following Pellegrino (2025), we measure product-market proximity, denoted by $\boldsymbol{\Psi}^\top \boldsymbol{\Psi}$, using the dataset developed by Hoberg and Phillips (2016). The measure is based on the cosine similarity of words appearing in the Business Description sections of 10-K filings. Hoberg and Phillips (2016) constructed a vocabulary of 61,146 words used by firms to describe the features and characteristics of their products. For each firm i, they construct

[4]We count Barclays Global Investors, whose index business BlackRock acquired in 2009, as part of BlackRock.

Figure 1: Average Ownership Weight, 2004–2019



Note: This figure plots the average of the ownership weights κ_{ij} over all pairs $i \neq j$ of sample firms in each year from 2004 to 2019. The series starts in 2004 because insiders have filed Forms 3, 4, and 5 electronically only since mid-2003, and 2004 is the first year whose June holdings reflect these filings.

a binary vector whose elements indicate whether the firm uses each word in its product description. Their analysis shows that cosine similarity effectively identifies industry groupings and predicts competitive relationships between firms, outperforming alternative industry classifications. Unlike the TNIC-3 industries of Hoberg and Phillips (2016), which keep only pairs above a similarity threshold, we use the similarity of every pair of sample firms.

The parameter α governs horizontal differentiation by mapping the cosine similarity between products into the magnitude of cross-price elasticities. We use the value of Pellegrino (2025), who sets α so that the model reproduces the inverse cross-price elasticity reported by Nevo (2001). In Table 2 of Pellegrino (2025), the own- and cross-price elasticities implied by the Generalized Hedonic Linear demand system are in line with IO estimates for automobiles (Berry et al., 1995), ready-to-eat cereal (Nevo, 2001), and computers (Goeree, 2008).

3.4 Technological Proximity and Spillovers

We measure technological proximity from patent data following Bloom et al. (2013) and Lucking et al. (2019). We use the DISCERN 2.0 dataset (Arora et al., 2024), which links U.S. publicly listed firms in Compustat to their patents.

Denote by T_i the vector of firm i's patent shares across technology classes, computed from the patents granted to firm i from year $t-2$ to $t+2$, the same five-year window used to select the sample. Because patents are granted with a lag after filing, grants in $t+1$ and $t+2$ largely reflect inventions made around t. Technology classes are main groups of the Cooperative Patent Classification, and a patent classified in several main groups counts once in each. Under the Jaffe measure (Jaffe, 1986), technological proximity is given by

$$\hat{\omega}_{ij} = \frac{T_i^\top T_j}{\left(T_i^\top T_i\right)^{1/2}\left(T_j^\top T_j\right)^{1/2}}.$$

As in Liu and Ma (2024), we normalize technological proximity so that each firm's weights sum to one,

$$\tilde{\omega}_{ij} = \frac{\hat{\omega}_{ij}}{\sum_{k\neq i}\hat{\omega}_{ik}}, \qquad j \neq i.$$

The normalization fixes each firm's total exposure to spillovers, so the strength of spillovers does not depend on how many firms the sample contains.

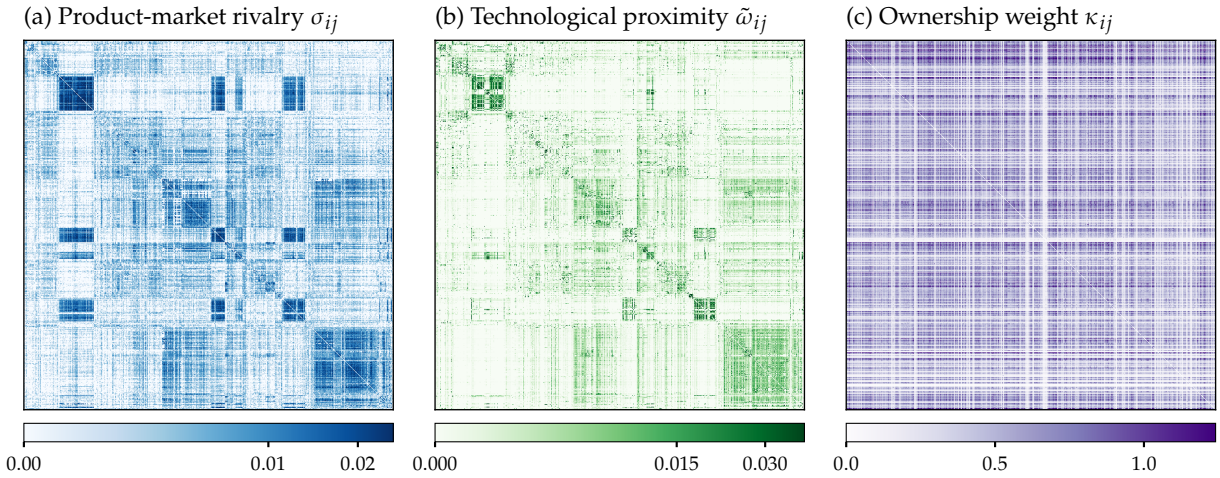
We set the coefficient β, which converts technological proximity into technology spillovers, to the estimate of Hopenhayn and Okumura (2026), $\beta = 0.02$. They estimate it as the association between firms' knowledge-capital growth and the knowledge capital of technologically proximate firms, with controls for own R&D intensity, own knowledge capital, exposure to product-market neighbors, and firm and year fixed effects. Section 4.7 reports the main results for $\beta = 0.01$ and $\beta = 0.03$.

3.5 Overlap of the Networks

The product-market rivalry and technological spillover networks overlap only partly. Across firm pairs in 2019, the correlation between product-market rivalry σ_{ij} and technological proximity $\tilde{\omega}_{ij}$ is 0.29, and 32 percent of the pairs in the top 5 percent by product-market rivalry are also in the top 5 percent by technological proximity. Ownership weights are nearly uncorrelated with both, with correlations of -0.03 with product-market rivalry and 0.01 with technological proximity. The business-stealing and technology-spillover externalities that common owners internalize therefore arise on partly different firm pairs.

Figure 2 plots the three networks with the firms sorted by industry. Product-market rivalry and technological proximity form dense blocks because both are concentrated within industries. Pairs of firms in the same four-digit NAICS industry (97 industries) make up only 3.8 percent of pairs but account for 16 percent of product-market rivalry and 22 percent of technological proximity. Ownership weights show no such blocks, and

Figure 2: Product-Market Rivalry, Technological Proximity, and Ownership Weights, 2019



Note: Each panel plots the off-diagonal entries of $\mathbf{\Sigma}$, $\widetilde{\mathbf{\Omega}}$, and $\boldsymbol{K}$ for the 638 sample firms in 2019. Row i shows firm i's rivalry with, spillovers from, and ownership weight on each other firm. Firms are sorted by six-digit NAICS code and, within a code, by sales.

pairs in the same industry account for 3.7 percent of them, about their share of pairs. Ownership weights instead form horizontal stripes, because the weight that a firm places on other firms depends mainly on its own owners. A firm whose votes are held mainly by diversified investors places high weights on all other firms, while a firm with a large insider or blockholder stake places low weights. The mean weight that each firm places on other firms accounts for 59 percent of the variance of ownership weights across pairs.

3.6 Knowledge Capital Distribution and Labor-Augmentation Efficiency

We recover quantities $\boldsymbol{q}_t$, labor-augmentation efficiency ζ/L, and knowledge capital z_t for 2019 from observed profits and production costs. The recovered z_t is a model-implied measure of knowledge capital.

The firm-level profit $\pi_{i,t}$ is measured as revenue (revt) minus cost of goods sold (cogs) in the Compustat database. Given product-market rivalry σ_{ij} and ownership weights κ_{ij}, we recover the quantity vector $\boldsymbol{q}_t$ so that it satisfies the following profit identity for each firm implied by the model:

$$\pi_{i,t} = \sum_j \kappa_{ij}\sigma_{ij}q_{i,t}q_{j,t}$$

This is a system of nonlinear equations in $\boldsymbol{q}_t$. Since $\kappa_{ii} = \sigma_{ii} = 1$, each equation can be

written as

$$\pi_{i,t} = q_{i,t}^2 + q_{i,t} \sum_{j \neq i} \kappa_{ij} \sigma_{ij} q_{j,t}.$$

We solve this system by iterating on the positive root of each equation,

$$q_{i,t} = \frac{-\sum_{j \neq i} \kappa_{ij} \sigma_{ij} q_{j,t} + \sqrt{\left(\sum_{j \neq i} \kappa_{ij} \sigma_{ij} q_{j,t}\right)^2 + 4\pi_{i,t}}}{2},$$

until the quantities converge. This procedure recovers quantities in the model's units. We then identify ζ/L using the following relationship in our model:

$$\frac{\zeta}{L} \boldsymbol{q}_t^\top \boldsymbol{J} \boldsymbol{q}_t = \text{total production cost at time } t.$$

Total production cost is the sum of cost of goods sold over the sample firms. Cost of goods sold is the data counterpart of the model's labor cost of production. In the model, ζ and L cannot be separately identified; only their ratio matters for equilibrium allocations. Finally, we recover the knowledge capital of firms $\boldsymbol{z}_t$ by inverting the static equilibrium condition (20):

$$\boldsymbol{z}_t = \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{K} \circ \boldsymbol{\Sigma}\right) \boldsymbol{q}_t.$$

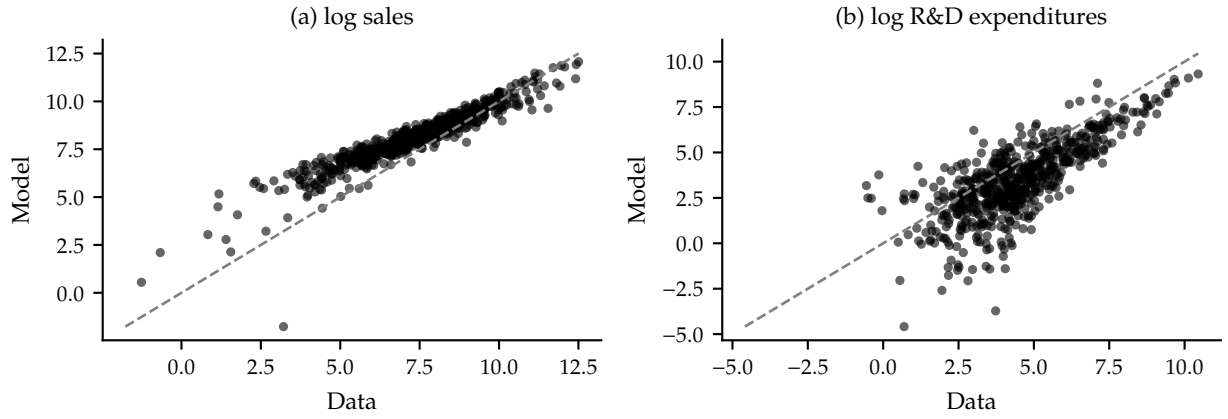
These conditions use the measured ownership weights because production ownership in the observed economy is $\boldsymbol{K}^{\mathrm{Q}} = \boldsymbol{K}$. The recovered knowledge capital is therefore the state of the observed economy, from which every counterfactual in Section 2.4 starts.

3.7 Discount Rate, R&D Efficiency, and Obsolescence

The remaining parameters are the discount rate ρ, the R&D efficiency μ, and the obsolescence rate δ. We set ρ to 10 percent. In the calibration, ρ acts mainly as the rate at which firms discount the returns to R&D, and surveys and conference calls show that firms evaluate investments at discount rates of about this size (Poterba and Summers, 1995; Gormsen and Huber, 2025). Section 4.7 reports results for ρ of 5 and 7.5 percent.

Holding ρ fixed, we calibrate μ and δ jointly to match two aggregate moments. First, the model's R&D expenditure share, the ratio of R&D expenditure $\sum_i x_{i,t}^2$ to labor payments plus the gross profits of the sample firms, equals R&D performed by businesses in 2019 as a share of business-sector gross value added (3.03 percent). Labor payments plus gross profits correspond to value added, so we compare the model's ratio with business-sector value added rather than GDP, which also includes the government, household, and nonprofit

Figure 3: Model and Data: Firm-Level Sales and R&D



Note: Each point is a firm in 2019. The horizontal axis is the log of the value in Compustat and the vertical axis the log of the value in the model; the dashed line is the 45-degree line. Panel (a) includes all 638 sample firms, and panel (b) includes the 612 firms with positive R&D expenditure in the data.

sectors. Second, the model's current growth rate of output at the observed state equals the average annual growth rate of U.S. real GDP per capita from 2004 to 2019 (1.165 percent), the period over which we measure the rise in common ownership. We use per-capita growth because Y_t is real output and the model holds the set of firms and the labor force fixed. Both targets are economy-wide values, so the calibration takes the structure of the sample as a miniature of the economy and matches the economy's R&D intensity and growth. Because the sample firms are more R&D-intensive than the business sector, their R&D in the model is below their reported R&D (Figure 3). R&D is from the National Patterns of R&D Resources of the National Center for Science and Engineering Statistics, and value added and GDP are from the Bureau of Economic Analysis.

3.8 Goodness of Fit

Figure 3 compares firm-level sales and R&D expenditures in the model and the data. The model matches firm-level profits by construction, so the figure omits them. Sales follow from the recovered quantities through the demand system, and R&D is the equilibrium policy of the dynamic game, which the calibration disciplines only through its aggregate. Neither is targeted at the firm level, and the correlations between the model and the data are 0.94 for log sales and 0.75 for log R&D expenditures.

Table 2: Growth Decomposition

Component	Contribution to growth in 2019 (%)
Technology spillovers	3.105
R&D	0.771
Obsolescence	−2.711
Total	1.165

Note: Contributions to output growth are decomposed using equations (32) to (34). Contributions are evaluated at the observed state.

4 Quantitative Results

This section solves the model at the observed state and quantifies how common ownership affects R&D, growth, welfare, and the value of firms. Growth rates refer to the current growth rate of output at the observed state (Section 4.2). We first describe the solution method and decompose growth at the observed state. We then report the main counterfactual, which changes ownership in R&D decisions, examine the mechanism, separate the production and R&D margins, study R&D policy and the heterogeneity of returns to R&D, and check robustness. All results use the interior linear solution of the model (Section B.2).

4.1 Solution Method

Because the static block has a closed form, the dynamic game reduces to the stacked Riccati equations (27), and welfare and the value of firms follow from Lyapunov equations. We solve these equations with the method of Hopenhayn and Okumura (2026) (Section B.1). With 638 firms, the matrices $\boldsymbol{X}^i$ contain $n^3 \simeq 2.6 \times 10^8$ coefficients, and each update costs $O\left(n^4\right)$ operations. The equilibrium solves in 19 seconds on a laptop. Table A.2 compares this scale with models that discretize each firm's productivity.

4.2 Growth Decomposition

The growth rate of output decomposes into contributions from technology spillovers, R&D, and the obsolescence of knowledge capital.

Using equations (26) and (28), the growth rate of output is[5]

$$\frac{d \log Y_t}{dt} = \frac{z_t^\top \left(\boldsymbol{Q}\boldsymbol{\Phi} + \boldsymbol{\Phi}^\top \boldsymbol{Q}\right) z_t}{z_t^\top \boldsymbol{Q} z_t}$$

[5]Section A.9 derives the corresponding process under the stochastic extension.

Because the technology transition matrix $\boldsymbol{\Phi}$ consists of the exogenous technological spillover network $\boldsymbol{\Omega}$, the endogenous R&D term $\mu^2\widetilde{\boldsymbol{X}}$, and obsolescence $-\delta\boldsymbol{I}$, we can further decompose this term

$$\boldsymbol{Q}\boldsymbol{\Phi} + \boldsymbol{\Phi}^\top\boldsymbol{Q} = \left(\boldsymbol{Q}\boldsymbol{\Omega} + \boldsymbol{\Omega}^\top\boldsymbol{Q}\right) + \mu^2\left(\boldsymbol{Q}\widetilde{\boldsymbol{X}} + \widetilde{\boldsymbol{X}}^\top\boldsymbol{Q}\right) - 2\delta\boldsymbol{Q}$$

We define the contributions of technology spillovers, R&D, and obsolescence to the growth rate of output as

$$\frac{d\log Y_t|_{\text{spillover}}}{dt} = \frac{\boldsymbol{z}_t^\top\left(\boldsymbol{Q}\boldsymbol{\Omega} + \boldsymbol{\Omega}^\top\boldsymbol{Q}\right)\boldsymbol{z}_t}{\boldsymbol{z}_t^\top\boldsymbol{Q}\boldsymbol{z}_t} \tag{32}$$

$$\frac{d\log Y_t|_{\text{R\&D}}}{dt} = \mu^2\frac{\boldsymbol{z}_t^\top\left(\boldsymbol{Q}\widetilde{\boldsymbol{X}} + \widetilde{\boldsymbol{X}}^\top\boldsymbol{Q}\right)\boldsymbol{z}_t}{\boldsymbol{z}_t^\top\boldsymbol{Q}\boldsymbol{z}_t} \tag{33}$$

$$\frac{d\log Y_t|_{\text{obsolescence}}}{dt} = -2\delta \tag{34}$$

Table 2 reports each contribution at the observed state. In the main counterfactual, ownership changes only R&D decisions, so the change in growth works entirely through the R&D term (33). The spillover term is proportional to β, and the obsolescence term equals -2δ, where δ is calibrated jointly with μ so that the three terms add up to observed growth. The levels of these two terms therefore reflect the values of these parameters.

4.3 Common Ownership in R&D Decisions

The main counterfactual raises R&D ownership from its 2004 intensity, $\left(\boldsymbol{K}^Q, \boldsymbol{K}^R\right) = (\boldsymbol{K}, \boldsymbol{K}_{2004})$, to the observed economy, $(\boldsymbol{K}, \boldsymbol{K})$ (Section 2.4.1). Table 3 reports R&D, the growth rate of output, social welfare, and the total value of firms.

Raising R&D ownership from $\boldsymbol{K}_{2004}$ to $\boldsymbol{K}$ lowers the growth rate of output by 0.087 percentage points, R&D expenditure by 21.2 percent, and social welfare by 0.32 percent. The fall in growth equals 11 percent of the 0.771 percentage points that the R&D term contributes in the observed economy (Table 2). This share is about half the fall in R&D expenditure because expenditure is quadratic in effort, which drives the accumulation of knowledge capital, so a given proportional fall in effort lowers expenditure by about twice as much.

Social welfare falls by less than slower growth alone would imply because lower R&D expenditure frees resources for consumption. Current consumption rises by 0.66 percent, but slower growth lowers future consumption by more, so lifetime consumption, our

Table 3: Common Ownership in R&D Decisions

	$\boldsymbol{K}^R = \boldsymbol{K}_{2004}$	$\boldsymbol{K}^R = \boldsymbol{K}$	Change
R&D (% of value added)	3.84	3.03	−21.2
Growth rate (%)	1.252	1.165	−0.087
Social welfare (index)	100.00	99.68	−0.32
Total value of firms (index)	100.00	101.08	+1.08

Note: Both columns set production ownership to its observed level, $\boldsymbol{K}^Q = \boldsymbol{K}$, and start from the knowledge capital of the observed economy. R&D is R&D expenditure as a percent of value added (labor payments plus the gross profits of the sample firms), the ratio targeted in the calibration. Because the two columns share production ownership and knowledge capital, they have the same value added, so the percent change in the R&D ratio equals that in R&D expenditure. Social welfare and the total value of firms are indexed to 100 in the first column. Changes are in percent, except the growth rate in percentage points.

measure of social welfare, falls. The total value of firms rises by 1.08 percent, and Section 4.4 traces the divergence between firm value and welfare to the internalization of business stealing.

4.4 Business Stealing, Technology Spillovers, and the Structure of Ownership

Table 4 reports three exercises that trace the main counterfactual to the forces behind it, to the placement of ownership across firms, and to ownership links within and across industries.

4.4.1 Business Stealing and Technology Spillovers

Common ownership leads firms to internalize two R&D externalities with opposite signs. To show the direction of each force, we repeat the main counterfactual in two economies that shut down one of them, holding all other parameters fixed. The business-stealing economy removes technology spillovers, $\boldsymbol{\Omega} = \mathbf{0}$. The technology-spillover economy removes cross-product substitution, $\boldsymbol{\Sigma} = \boldsymbol{I}$, and the interaction through the common real wage, $\zeta/L = 0$. Because the two forces interact in equilibrium, the effects in the two economies need not add up to the effect in the full model.

In the full model, raising R&D ownership lowers R&D, as it does in the business-stealing economy, so in the measured networks the internalization of business stealing dominates the effect on R&D. A calculation that holds R&D policies fixed at the observed state shows why the spillover force is weak.[6] Raising R&D ownership lowers the value that owners

[6]The calculation is a first-order approximation that weights firms by knowledge capital. Section B.3 gives

Table 4: Business Stealing, Technology Spillovers, and the Structure of Ownership

	R&D expenditure (%)	Growth rate (pp)	Social welfare (%)	Total value of firms (%)
Full model	−21.2	−0.087	−0.32	+1.08
Business-stealing economy	−20.0	−0.080	−0.16	+0.67
Technology-spillover economy	+18.8	+0.246	+2.22	+0.75
Uniform R&D ownership ($\bar{\boldsymbol{K}}$)	−10.6	−0.028	−0.11	+1.31
Rise only within industries	−1.9	−0.007	−0.02	+0.10
Rise only across industries	−19.5	−0.080	−0.29	+1.01

Note: The first three rows raise R&D ownership from $\boldsymbol{K}_{2004}$ to $\boldsymbol{K}$ in the full model and in the two restricted economies. The business-stealing economy sets $\boldsymbol{\Omega} = \boldsymbol{0}$, and the technology-spillover economy sets $\boldsymbol{\Sigma} = \boldsymbol{I}$ and $\zeta/L = 0$. The fourth row raises R&D ownership from $\boldsymbol{K}_{2004}$ to the uniform matrix $\bar{\boldsymbol{K}}$ in the full model. The last two rows raise R&D ownership from $\boldsymbol{K}_{2004}$ to $\boldsymbol{K}$ only between firms in the same NAICS four-digit industry or only between firms in different industries. For each outcome, the sum of the two rows differs from the full-model effect by at most 2.8 percent of that effect. With NAICS three-digit industries, links across industries account for 87 percent of the growth effect. All rows set production ownership to $\boldsymbol{K}^{\mathrm{Q}} = \boldsymbol{K}$ and start from the knowledge capital of the observed economy.

attach to a unit of a firm's knowledge through business stealing and raises it through spillovers. Because spillovers reach other firms only gradually, at rate β, the rise through spillovers would be only 25 percent of the fall through business stealing even if the spilled knowledge took no business. The knowledge that spills over to another firm, however, also takes business from third firms in which the same owners hold stakes, so the rise is only 5 percent of the fall. In the technology-spillover economy, rivalry is absent and spilled knowledge takes no business, so the spillover force raises R&D.

Both forces raise the value of firms, but they move social welfare in opposite directions. In the business-stealing economy, raising R&D ownership raises the total value of firms by 0.67 percent and lowers social welfare by 0.16 percent. In the technology-spillover economy, it raises the total value of firms by 0.75 percent and social welfare by 2.22 percent. Common owners internalize the effects of R&D on the values of the firms in which they hold stakes. Technology spillovers raise both these values and social welfare, so internalizing them raises welfare. The profits that a firm's R&D takes from commonly owned rivals are a loss to their owners, but the gains that R&D brings to consumers and workers do not enter the owners' objective. Internalizing business stealing therefore lowers R&D, and because firms already invest too little in R&D, as Section 4.6 shows for the full model, it lowers welfare. In the full model the business-stealing force dominates the effect on welfare, so the main counterfactual lowers welfare while raising the value of firms.

its formulas and results with firms weighted equally or by R&D expenditure.

4.4.2 Placement of Ownership

The fourth row of Table 4 raises R&D ownership from $\boldsymbol{K}_{2004}$ to the uniform matrix $\bar{\boldsymbol{K}}$, which spreads the mean off-diagonal weight of $\boldsymbol{K}$ evenly across firm pairs, instead of $\boldsymbol{K}$ (Section 2.4.1). The placement of ownership weights across firms accounts for most of the growth effect. Raising R&D ownership to the uniform matrix $\bar{\boldsymbol{K}}$ lowers growth by only 0.028 percentage points, 68 percent less than raising it to $\boldsymbol{K}$. The two matrices differ in which firms carry large weights, not in how ownership links line up with product-market rivalry or technological proximity, because at the pair level the observed weights are nearly uncorrelated with both (correlations of -0.03 and 0.01; Section 3). The largest fifth of firms by knowledge capital accounts for 82 percent of R&D and places the largest weights on other firms' values (Table A.4), so the R&D-weighted mean weight on other firms is 0.59, compared with the unweighted mean of 0.42 that the uniform matrix gives every firm. Their R&D falls by 16.9 percent under the observed ownership but by only 4.6 percent under the uniform matrix, because spreading the weights evenly shifts internalization toward small firms that do little R&D.

Figure 4 shows how concentrated the fall in R&D is. The largest fifth of firms accounts for 66 percent of the fall in aggregate R&D under the observed ownership but only 36 percent of the smaller fall under the uniform matrix.

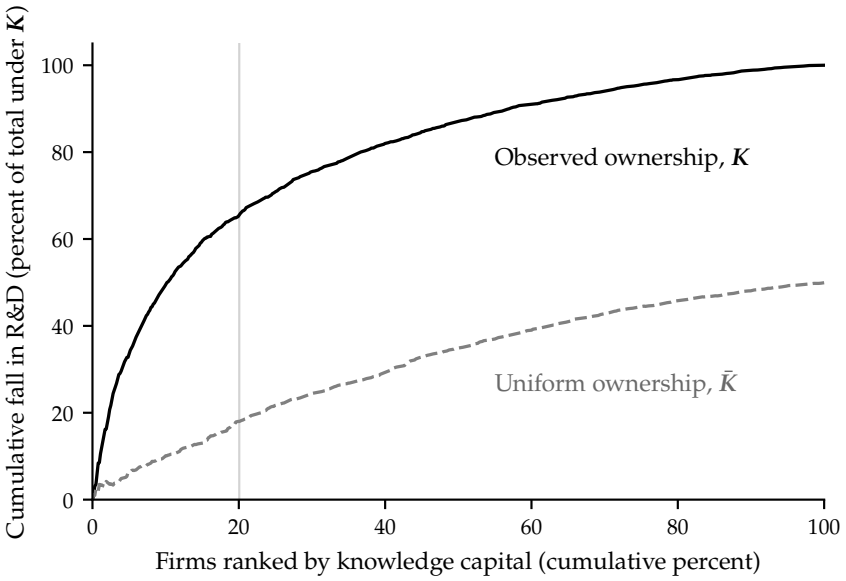
4.4.3 Ownership Links within and across Industries

Models in which firms interact strategically only within industries cannot capture ownership links across industries. Using the four-digit NAICS industries of Section 3.5, we split the rise in R&D ownership into links within and across industries. The rise is proportional to the ownership weights, so like them it is spread across firm pairs almost regardless of industry. Links between firms in different industries account for 96 percent of the rise and 92 percent of the growth effect (Table 4). Pairs in the same industry are closer in both networks, and per unit of the rise in weights, links within industries lower growth 2.2 times as much as links across industries. They carry too little of the rise, however, to account for much of the aggregate effect. The split changes only the ownership links, and product-market rivalry and technological proximity across industries stay as measured.

4.5 Production and R&D Ownership

The main counterfactual holds production ownership at its observed level, but common ownership also changes product-market conduct. To separate the two decision margins,

Figure 4: Where R&D Falls under Observed and Uniform Ownership



Note: Firms are ranked by knowledge capital in 2019, from largest to smallest. Each line cumulates the fall in R&D expenditure from raising R&D ownership from $\boldsymbol{K}_{2004}$ to the observed matrix $\boldsymbol{K}$ or to the uniform matrix $\bar{\boldsymbol{K}}$, as a percent of the total fall under $\boldsymbol{K}$. Production ownership is $\boldsymbol{K}^Q = \boldsymbol{K}$. The vertical line marks the largest fifth of firms.

Table 5 reports the four combinations $\left(\boldsymbol{K}^Q, \boldsymbol{K}^R\right) \in \{\boldsymbol{K}_{2004}, \boldsymbol{K}\}^2$. For an outcome y, the production margin is $y\left(\boldsymbol{K}, \boldsymbol{K}_{2004}\right) - y\left(\boldsymbol{K}_{2004}, \boldsymbol{K}_{2004}\right)$, the R&D margin is $y\left(\boldsymbol{K}_{2004}, \boldsymbol{K}\right) - y\left(\boldsymbol{K}_{2004}, \boldsymbol{K}_{2004}\right)$, and the interaction is the remainder of the total change $y\left(\boldsymbol{K}, \boldsymbol{K}\right) - y\left(\boldsymbol{K}_{2004}, \boldsymbol{K}_{2004}\right)$.

The two margins move growth in opposite directions. Softer product-market competition raises the return to a firm's own knowledge capital (Section A.4), so raising production ownership raises R&D expenditure by 10.1 percent and the growth rate by 0.036 percentage points. Unlike in the main counterfactual, not all of this change works through the R&D term. Production ownership also changes the matrix $\boldsymbol{Q}$ that maps knowledge capital into output, and with it the spillover term in equation (32), so 0.007 of the 0.036 percentage points come through the spillover term and 0.029 through the R&D term. Raising production ownership lowers social welfare by 0.95 percent, because softer competition lowers output, while it raises the total value of firms by 7.77 percent. Raising R&D ownership lowers R&D expenditure by 19.3 percent, the growth rate by 0.079 percentage points, and welfare by 0.35 percent. The interaction lowers growth by only 0.008 percentage points, so raising R&D ownership lowers growth by about as much with production ownership at its 2004 intensity (0.079 percentage points) as at its observed level in the main counterfactual (0.087 percentage points). Softer product-market competition thus accounts for 0.95 of the 1.27 percent welfare cost of common ownership even though it raises growth.

Table 5: Production and R&D Ownership

Panel A. Outcomes				
Production ownership $\boldsymbol{K}^Q$	$\boldsymbol{K}_{2004}$	$\boldsymbol{K}$	$\boldsymbol{K}_{2004}$	$\boldsymbol{K}$
R&D ownership $\boldsymbol{K}^R$	$\boldsymbol{K}_{2004}$	$\boldsymbol{K}_{2004}$	$\boldsymbol{K}$	$\boldsymbol{K}$
Output (index)	100.00	98.91	100.00	98.91
R&D (% of value added)	3.49	3.84	2.82	3.03
Growth rate (%)	1.216	1.252	1.137	1.165
Social welfare (index)	100.00	99.05	99.65	98.73
Firm value share (%)	28.1	30.6	28.5	31.0

Panel B. Decomposition of the total change				
	Production	R&D	Interaction	Total
R&D expenditure (%)	+10.1	−19.3	−4.0	−13.2
Growth rate (pp)	+0.036	−0.079	−0.008	−0.051
Social welfare (%)	−0.95	−0.35	+0.04	−1.27
Total value of firms (%)	+7.77	+0.95	+0.22	+8.94

Note: Output and social welfare are indexed to 100 in the first column. R&D in Panel A is R&D expenditure as a percent of value added (labor payments plus the gross profits of the sample firms), the ratio targeted in the calibration. Panel B reports percent changes in R&D expenditure and in the total value of firms. All changes in Panel B are measured from $(\boldsymbol{K}^Q, \boldsymbol{K}^R) = (\boldsymbol{K}_{2004}, \boldsymbol{K}_{2004})$, whereas those in Table 3 are measured from $(\boldsymbol{K}, \boldsymbol{K}_{2004})$.

Fully dispersed ownership, $\boldsymbol{K}^Q = \boldsymbol{K}^R = \boldsymbol{I}$, in which no firm places weight on other firms' values, yields a growth rate of 1.242 percent, above both the 1.216 percent with ownership in both decisions at its 2004 intensity and the observed economy's 1.165 percent.

4.6 R&D Policy

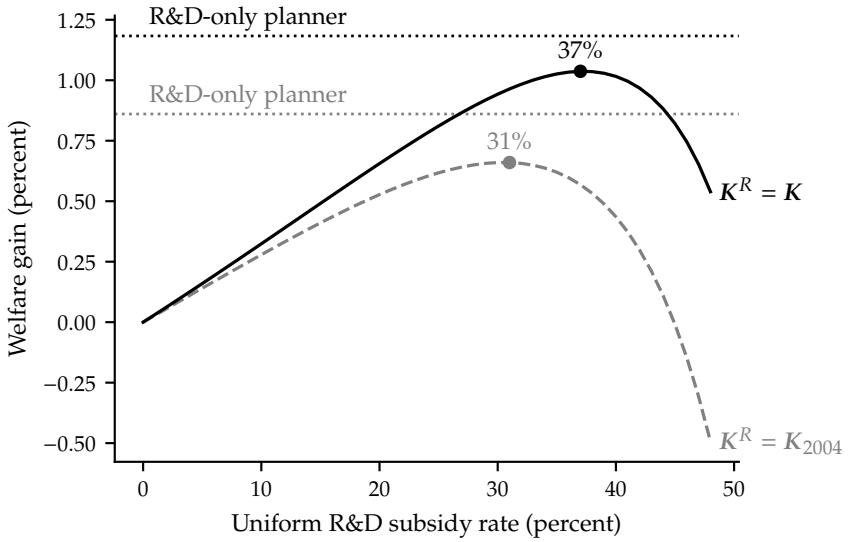
4.6.1 Uniform R&D Subsidy

We first analyze the welfare-maximizing uniform R&D subsidy. A subsidy rate s reduces each firm's private R&D cost by the factor $1 - s$, so the firm's R&D policy satisfies $\boldsymbol{x} = (1 - s)^{-1}\mu\widetilde{\boldsymbol{X}}\boldsymbol{z}$. The subsidy is financed by lump-sum taxes, so it changes private R&D incentives but not the resource cost of R&D, and household consumption subtracts the full cost $\boldsymbol{x}^\top\boldsymbol{x}$.

We compare the subsidy with the R&D-only planner of Section 2.4.2, which, like the subsidy, leaves product-market conduct unchanged. We solve the equilibrium over a grid of subsidy rates and report the welfare gain at each rate in the observed economy and in the economy with R&D ownership at its 2004 intensity (Figure 5).

The welfare-maximizing subsidy is positive in both economies, and the R&D-only

Figure 5: Uniform R&D Subsidy and the R&D-Only Planner



Note: Both curves set production ownership to $\boldsymbol{K}^Q = \boldsymbol{K}$. Welfare gains are consumption-equivalent changes relative to the equilibrium without a subsidy. Dots mark the welfare-maximizing rates, and dotted lines mark the welfare gain of the R&D-only planner in each economy. The subsidy rate varies in one-percentage-point steps from 0 to 48 percent, and welfare peaks inside this range in both economies.

planner spends 2.25 times as much on R&D as the equilibrium when R&D ownership is at its 2004 intensity and 2.85 times as much in the observed economy, so firms invest too little in R&D. Common ownership widens this gap. Because both economies set $\boldsymbol{K}^Q = \boldsymbol{K}$, the R&D-only planner's allocation is the same in the two, so the comparison reflects how far equilibrium R&D lies from a common benchmark. Raising R&D ownership from $\boldsymbol{K}_{2004}$ to $\boldsymbol{K}$ raises the planner's welfare gain from 0.86 to 1.18 percent, the welfare-maximizing subsidy from 31 to 37 percent, and the welfare gain from the subsidy from 0.66 to 1.04 percent. The subsidy attains 88 percent of the planner's gain in the observed economy, compared with 77 percent when R&D ownership is at its 2004 intensity. The share rises because common ownership widens the shortfall of equilibrium R&D from the planner's mainly in its overall scale, which a uniform subsidy can close. Raising all firms' R&D in the same proportion closes 76 and 88 percent of the squared distance between the equilibrium and planner's R&D vectors in the two economies, close to the shares of the planner's gain that the subsidy attains.

4.6.2 Heterogeneity in Private and Social Returns to R&D

The uniform subsidy falls short of the planner's gain because the gap between the social and private returns to R&D differs across firms, whereas the subsidy lowers every firm's

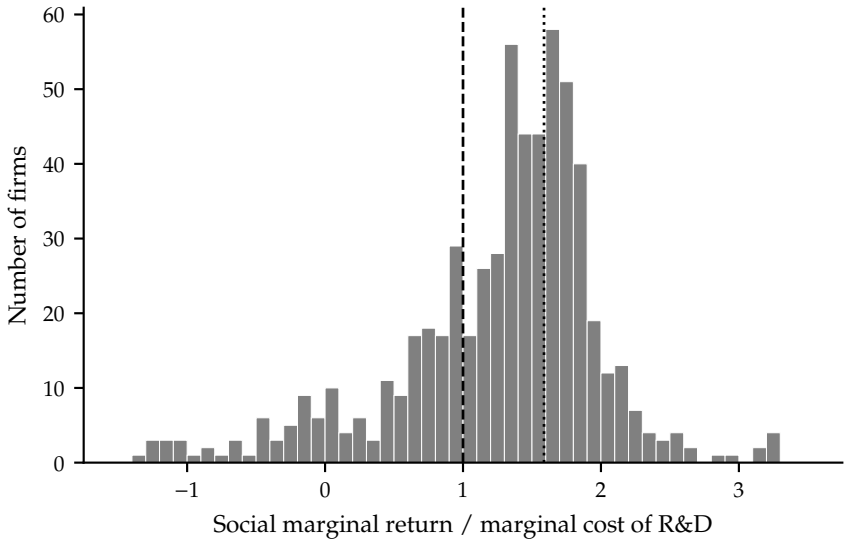
R&D cost by the same factor. It closes most of the shortfall in the overall scale of R&D but not in its allocation across firms. We therefore compare the two returns firm by firm. The private marginal return to firm i's R&D is the increase in the ownership-weighted value V^i that the firm maximizes from an additional unit of its R&D, $\mu\, dV^i(\boldsymbol{z})/dz_i = 2\mu \left(\boldsymbol{X}_i^i\right)^\top \boldsymbol{z}$, and the social marginal return is the increase in social welfare. Stacked across firms,

$$\text{Private marginal return to R\&D} = 2\mu\, \widetilde{\boldsymbol{X}}\, \boldsymbol{z},$$
$$\text{Social marginal return to R\&D} = \mu\, \frac{d}{d\boldsymbol{z}} V(\boldsymbol{z}) = 2\mu\, \boldsymbol{X}\, \boldsymbol{z}.$$

Because the private marginal return equals the marginal resource cost $2x_i$ at the firm's optimum, the ratio of the social to the private marginal return is the social return per unit of marginal cost. A ratio above one indicates that the firm underinvests at the margin, and a negative ratio indicates that its R&D lowers welfare at the margin. A subsidy at rate s aligns firm i's R&D incentive with its social return at the observed state when its ratio equals $1/(1-s)$. A uniform subsidy therefore under-rewards firms with higher ratios, which still invest too little at the margin, and over-rewards firms with lower ratios. Figure 6 shows the distribution of the ratio across the 630 firms with positive equilibrium R&D in the observed economy. The median ratio is 1.40, and 30 percent of firms have ratios below one, including 10 percent with negative ratios. The welfare-maximizing uniform subsidy aligns incentives at a ratio of 1.59. Firms above this ratio make up 37 percent of firms but 70 percent of equilibrium R&D expenditure, so the uniform subsidy under-rewards the firms that do most of the R&D.

To describe which firms the uniform subsidy over- or under-rewards, we regress each firm's ratio of social to private marginal return on its eigenvector centralities in the ownership, product-market rivalry, and technological spillover networks, controlling for its initial knowledge capital and using the firms that also report positive R&D in the data (Table 6). Firms central in the product-market rivalry network have lower ratios, and firms central in the technological spillover and ownership networks have higher ratios. The signs of the product-market rivalry and technological spillover centralities follow the externalities that separate the two returns. The profits that a firm's R&D takes from its rivals count toward its private return but not toward the social return, whereas the knowledge it spills over to other firms raises the social return but not its private return. Ownership centrality enters with a positive sign because, as in Section 4.4, the internalization of business stealing dominates. A firm that places large weight on other firms' values therefore has a lower private return relative to its social return, and such firms are among those that a uniform subsidy under-rewards. The firms that do most of the R&D are central in the ownership

Figure 6: Social Marginal Return to R&D per Unit of Marginal Cost



Note: The histogram shows, for the 630 firms with positive equilibrium R&D in the observed economy, the ratio of the social marginal return to R&D, $2\mu \boldsymbol{X}\boldsymbol{z}$, to its marginal cost, which equals the private marginal return $2\mu \widetilde{\boldsymbol{X}}\boldsymbol{z}$ at the firm's optimum. The dashed line marks a ratio of one. The dotted line marks $1/(1-s)$ at the welfare-maximizing uniform subsidy s of 37 percent. The figure omits 23 firms with ratios below −1.5 or above 3.5. Returns are computed at the observed state.

and technological spillover networks. Weighted by equilibrium R&D expenditure, the mean ownership centrality is 0.54 rather than 0.38, and the mean technological spillover centrality is 0.51 rather than 0.22, whereas the weighted mean product-market rivalry centrality is below its unweighted mean.

4.7 Robustness

Table 7 repeats the main counterfactual under alternative values of the technology-spillover coefficient β, the product-substitutability parameter α, and the discount rate ρ, and with ownership measured from 13F filings alone. The 13F specification uses the institutional holdings of Backus et al. (2021) for 2017 and measures the rise in ownership from 1999, with its own sample, knowledge capital, and calibration targets. Because 13F filings omit insiders and other owners that are not institutional investors, it corrects the denominator of κ_{ij} for missing owners following Ederer and Pellegrino (2026), who assume that unobserved holdings resemble observed ones. R&D efficiency μ and obsolescence δ are recalibrated in each specification.

In every specification, raising R&D ownership lowers R&D expenditure, growth, and social welfare. In 1999 the average ownership weight was 0.20 times its 2017 level, compared

Table 6: Network Positions and the Ratio of Social to Private Returns to R&D

	Social / private return to R&D
Initial knowledge capital (thousands)	-1.87^{***}
	(0.48)
Product-market rivalry centrality	-2.21^{***}
	(0.30)
Technological spillover centrality	2.28^{***}
	(0.24)
Ownership centrality	2.78^{***}
	(0.29)
Observations	605
R^2	0.39

Note: Firm-level OLS regression at the observed state, with heteroskedasticity-robust (HC1) standard errors in parentheses; ***, **, and * denote significance at the 1, 5, and 10 percent levels. The dependent variable is the ratio of the social to the private marginal return in levels, because the ratio can be negative. The sample consists of the firms with positive R&D expenditure in the data and a positive private marginal return. Initial knowledge capital enters in levels, and its coefficient is reported per thousand units. Each centrality is the absolute value of the eigenvector associated with the eigenvalue with the largest real part, divided by its maximum, computed from $\boldsymbol{K}$, $\boldsymbol{\Sigma}$, or $\widetilde{\boldsymbol{\Omega}}$ with zero diagonal. Ownership and product-market rivalry centralities use the right eigenvector. Because each row of $\widetilde{\boldsymbol{\Omega}}$ sums to one, its right eigenvector is constant across firms, so technological spillover centrality uses the left eigenvector, which measures a firm's importance as a source of spillovers.

with a ratio of 0.48 between 2004 and 2019 in the baseline, so the 13F specification implies larger falls in R&D and growth. A larger β raises the growth and welfare losses, because each unit of R&D that firms forgo also forgoes more knowledge that would have spread to other firms. A larger α makes products more substitutable and strengthens business stealing, so R&D and growth fall by more, while the welfare loss becomes slightly smaller. A lower discount rate ρ raises the value that firms place on future returns to R&D, so matching the R&D expenditure share requires a lower R&D efficiency μ. Each unit of R&D then adds less knowledge capital, and a similar proportional fall in R&D lowers growth by less. The welfare loss is nevertheless larger, because households discount the future consumption lost to slower growth less heavily. Relative to lower discount rates, the baseline value $\rho = 0.10$ thus gives a larger growth effect and a smaller welfare loss.

5 Alternative Models of Corporate Governance

The baseline assumes that each owner's influence on a firm's decisions is proportional to the owner's voting share. Because the effect of common ownership depends on how

Table 7: Robustness of the Main Counterfactual

Specification	R&D expenditure (%)	Growth rate (pp)	Social welfare (%)
Baseline	−21.2	−0.087	−0.32
$\beta = 0.01$	−20.8	−0.080	−0.25
$\beta = 0.03$	−21.1	−0.091	−0.40
$\alpha = 0.08$	−19.5	−0.083	−0.35
$\alpha = 0.16$	−22.7	−0.090	−0.29
$\rho = 0.05$	−20.1	−0.040	−0.52
$\rho = 0.075$	−21.1	−0.065	−0.38
13F ownership, 1999–2017	−29.6	−0.111	−0.32

Note: Each row except the last reports the change from raising R&D ownership from $\boldsymbol{K}_{2004}$ to $\boldsymbol{K}$ with production ownership at $\boldsymbol{K}^{Q} = \boldsymbol{K}$. The last row does the same with the 13F matrix for 2017, scaled to its 1999 mean intensity for the starting point. The baseline sets $\beta = 0.02$, $\alpha = 0.12$, and $\rho = 0.10$.

shareholdings translate into the weights that managers place on other firms' values, we examine alternative assumptions about this mapping. Following Ederer and Pellegrino (2026), we consider three models: (1) super-proportional influence of large owners, (2) a blockholder threshold, and (3) governance frictions and managerial entrenchment, the last in two versions. Each model builds its weights from the same voting shares v_{io} and cash-flow shares s_{io} and nests the baseline as a special case. We keep the baseline knowledge capital and parameters, so differences across models reflect only the mapping from shareholdings to weights. Each model yields its own ownership matrix $\boldsymbol{K}$. We construct its base-year matrix $\boldsymbol{K}_{2004}$ as in the baseline, scaling the off-diagonal weights of $\boldsymbol{K}$ by the model's own ratio of the 2004 to the 2019 mean off-diagonal weight, and we repeat the main counterfactual of Section 4.3 under each model.

5.1 Super-Proportional Influence of Large Owners

Following Gilje et al. (2020) and Backus et al. (2021), we assume that firm i weights the portfolio value $\tilde{V}^{o}$ of owner o by $\chi_{io} v_{io}$, where χ_{io} captures the greater influence of large owners. Then, the influence-adjusted ownership weight $\tilde{\kappa}_{ij}$ is

$$\tilde{\kappa}_{ij} = \frac{\sum_{o=1}^{n_o} \chi_{io}\, v_{io}\, s_{jo}}{\sum_{o=1}^{n_o} \chi_{io}\, v_{io}\, s_{io}}, \quad i \neq j.$$

Setting $\chi_{io} = 1$ nests the baseline. Following Ederer and Pellegrino (2026), we set $\chi_{io} = \sqrt{v_{io}}$, which amplifies the impact of large owners on firm decisions.

5.2 Blockholder Threshold

An alternative way to model the disproportionate control of large owners is to assume that only owners who hold more than a given fraction of a firm's voting rights influence that firm's decisions. Accordingly, firm i weights the portfolio value of owner o by v_{io} if o is a blockholder of firm i and by zero otherwise. We set the threshold at 5 percent, the ownership level above which owners must disclose their stakes on Schedule 13D or 13G. Define $b_{io} \equiv \mathbb{1}\{v_{io} > 0.05\}$. The resulting ownership weight is

$$\tilde{\kappa}_{ij} = \frac{\sum_{o=1}^{n_o} b_{io}\, v_{io}\, s_{jo}}{\sum_{o=1}^{n_o} b_{io}\, v_{io}\, s_{io}}, \quad i \neq j.$$

A firm without any blockholder maximizes its own value, so $\tilde{\kappa}_{ij} = 0$ for all $j \neq i$.

5.3 Governance Frictions and Managerial Entrenchment

Corporate managers may not fully maximize shareholder value due to governance frictions and entrenchment. Azar and Ribeiro (2022) construct a structural model of industrial organization and estimate it using airline industry data to quantify ownership weights. They propose a flexible objective function in which the manager of firm i discounts the profits of other firms by a mitigation factor τ_i relative to proportional influence. In our dynamic setting, firm i then maximizes

$$\hat{V}^i + \tau_i \sum_{j \neq i} \kappa_{ij} \hat{V}^j.$$

This objective multiplies firm i's off-diagonal ownership weights by τ_i. We use their baseline estimates, which account for overlapping ownership across industries, and analyze two versions. First, we set $\tau_i = 0.3771$ for all firms, their estimate when the mitigation factor is common to all firms. This version scales all off-diagonal weights equally and preserves the relative weights across links. Second, we let τ_i vary across firms according to their structural specification,

$$\tau_i = \frac{e^{\theta_0}\, \mathrm{IHHI}_i}{1 + e^{\theta_0}\, \mathrm{IHHI}_i}$$

with $\theta_0 = 3.2454$, where $\mathrm{IHHI}_i = \sum_{o=1}^{n_o} v_{io} s_{io}$ is the denominator of κ_{ij}. It equals the investor Herfindahl concentration index $\sum_{o=1}^{n_o} s_{io}^2$ when voting and cash-flow rights coincide. Because τ_i increases with IHHI_i, managers of firms with more concentrated ownership discount other firms' values less. We compute IHHI_i from each year's holdings.

Table 8: Alternative Models of Corporate Governance

	Proportional influence	Super-proportional influence	Blockholder threshold	Governance frictions (uniform)	Governance frictions (firm-specific)
Mean weight, 2004	0.20	0.17	0.11	0.08	0.08
Mean weight, 2019	0.42	0.42	0.43	0.16	0.20
R&D expenditure (%)	−21.2	−22.8	−29.1	−7.4	−9.7
Growth rate (pp)	−0.087	−0.096	−0.130	−0.029	−0.036
Social welfare (%)	−0.32	−0.35	−0.44	−0.09	−0.11

Note: The first two rows report the mean off-diagonal weight of each model's base-year matrix $\boldsymbol{K}_{2004}$ and observed matrix $\boldsymbol{K}$. The remaining rows report the change from raising R&D ownership from $\boldsymbol{K}_{2004}$ to $\boldsymbol{K}$, with production ownership at $\boldsymbol{K}^{Q} = \boldsymbol{K}$, under the corresponding mapping from shareholdings to ownership weights. Proportional influence is the baseline.

5.4 Results

Table 8 reports the main counterfactual under each governance model. In every model, raising R&D ownership lowers R&D expenditure, growth, and social welfare, and the fall in the growth rate ranges from 0.029 to 0.130 percentage points. The size of the effect follows how much each model raises the mean ownership weight between 2004 and 2019. The three models without governance frictions give similar mean weights in 2019, but super-proportional influence and, even more, the blockholder threshold give lower weights in 2004, so they imply a larger rise in common ownership and a larger fall in growth. Governance frictions scale down all weights, and with them the rise in the mean weight, so they yield the smallest effects.

6 Conclusion

We develop an endogenous growth model that incorporates three inter-firm networks: ownership, product-market rivalry, and technological spillovers. We quantify the model with data on 638 publicly traded U.S. firms with patents. Relative to an economy in which only R&D decisions reflect 2004 ownership intensity, the 2019 economy has 21.2 percent less R&D, a 0.087 percentage point lower growth rate, and 0.32 percent lower social welfare, but a 1.08 percent higher total value of firms. R&D is lower because, in R&D decisions, the internalization of business stealing outweighs that of technology spillovers. Firm value and welfare move apart because common owners account for the effects of R&D on the values of the firms in which they hold stakes but not for its effects on consumers and workers. Because firms already invest too little in R&D, the fall in R&D lowers welfare. By

contrast, raising the ownership weights in production decisions from their 2004 intensity softens competition, which raises growth but lowers welfare, so the growth cost of common ownership arises from R&D decisions.

The framework also applies beyond U.S. common ownership. Family-controlled business groups account for a large share of economic activity in South Korea, where they are called chaebols, and in other East and Southeast Asian economies. In Japan, the Allied occupation dissolved the conglomerates called zaibatsu after World War II. The framework can assess how such groups, and their dissolution, affect product-market competition, R&D activity, and economic growth. It can also quantify the net effect of foreign direct investment (FDI) in developing economies, where policymakers weigh market-share losses to foreign entrants against gains from technology spillovers. The product-market rivalry and technological spillover networks measure these losses and gains for each domestic firm. Finally, treating technology licenses as claims on the product-market profits that the licensed technology generates lets licenses enter the framework in the same way as ownership shares, so the framework can quantify how licensing shapes product-market competition, R&D investment, and aggregate growth.

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Appendix

Appendix A Model Derivations and Proofs

A.1 Derivation of Static Equilibrium

Inserting the optimality conditions for technology choice (17) and (18) into the optimality condition for quantity choice (16), we obtain

$$z_{i,t} = 2\sqrt{\zeta \frac{w_t}{P_t}} + \sum_j \left(\sigma_{ij} + \sigma_{ij}\kappa_{ij}^Q\right) q_{j,t}$$

Inserting the expression for real wage (19), we obtain

$$z_{i,t} = \sum_j \left(2\frac{\zeta}{L} + \sigma_{ij} + \sigma_{ij}\kappa_{ij}^Q\right) q_{j,t}.$$

In vector form,

$$\boldsymbol{z}_t = \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{K}^Q \circ \boldsymbol{\Sigma}\right) \boldsymbol{q}_t.$$

Therefore, quantities are linear functions of knowledge capital,

$$\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t$$

where

$$\boldsymbol{N} \equiv \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{K}^Q \circ \boldsymbol{\Sigma}\right)^{-1}.$$

A.2 Derivation of Quadratic Form of Ownership-Weighted Profits

Substituting the first-order condition with respect to q_i into the expression for the profit (15), we obtain

$$\pi_{i,t} = \sum_j \kappa_{ij}^Q \sigma_{ij} q_{i,t} q_{j,t}$$

Therefore, the profit flow in firm i's R&D objective, weighted by the R&D ownership weights, is

$$\sum_j \kappa_{ij}^R \pi_{j,t} = \sum_h \kappa_{ih}^R \sum_j \kappa_{hj}^Q \sigma_{hj} q_{h,t} q_{j,t} \tag{35}$$

Let $\boldsymbol{N}_i$ denote the i-th row of $\boldsymbol{N}$. From equation (20), we obtain

$$q_{h,t} q_{j,t} = \boldsymbol{z}_t^\top \left(\frac{\boldsymbol{N}_h^\top \boldsymbol{N}_j + \boldsymbol{N}_j^\top \boldsymbol{N}_h}{2} \right) \boldsymbol{z}_t$$

Inserting the above equation into equation (35), we express the weighted profit flow as a quadratic form of knowledge capital $\boldsymbol{z}_t$,

$$\sum_j \kappa_{ij}^R \pi_{j,t} = \boldsymbol{z}_t^\top \boldsymbol{Q}^i \boldsymbol{z}_t$$

where

$$\boldsymbol{Q}^i \equiv \sum_h \kappa_{ih}^R \sum_j \kappa_{hj}^Q \sigma_{hj} \frac{\boldsymbol{N}_h^\top \boldsymbol{N}_j + \boldsymbol{N}_j^\top \boldsymbol{N}_h}{2}$$

A.3 Derivation of Output

The derivation follows Hopenhayn and Okumura (2026), with the production ownership weights $\boldsymbol{K}^Q$ added. Let $\mathbf{1}$ denote an $n \times 1$ vector with all elements equal to 1. In vector form, equation (18) is

$$\boldsymbol{b}_t = \boldsymbol{z}_t - \sqrt{\zeta \frac{w_t}{P_t}} \mathbf{1}$$

Inserting equation (19), we obtain

$$\begin{aligned} \boldsymbol{b}_t &= \boldsymbol{z}_t - \frac{\zeta}{L} \left(\sum_i q_{i,t} \right) \mathbf{1} \\ &= \boldsymbol{z}_t - \frac{\zeta}{L} \boldsymbol{J} \boldsymbol{q}_t. \end{aligned}$$

From equation (20), we obtain

$$\boldsymbol{b}_t = \left(\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{K}^Q \circ \boldsymbol{\Sigma} \right) \boldsymbol{q}_t$$

Therefore, from equation (4), the total output is given by

$$\begin{aligned} Y_t &= \boldsymbol{q}_t^\top \left(\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{K}^Q \circ \boldsymbol{\Sigma} \right) \boldsymbol{q}_t - \frac{1}{2} \boldsymbol{q}_t^\top \boldsymbol{\Sigma} \boldsymbol{q}_t \\ &= \boldsymbol{q}_t^\top \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{K}^Q \circ \boldsymbol{\Sigma} \right) \boldsymbol{q}_t \\ &= \boldsymbol{z}_t^\top \boldsymbol{N}^\top \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{K}^Q \circ \boldsymbol{\Sigma} \right) \boldsymbol{N} \boldsymbol{z}_t \\ &= \frac{1}{2} \boldsymbol{z}_t^\top \boldsymbol{N}^\top \left(2 \frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{\Sigma} \circ \left(\boldsymbol{K}^Q + \left(\boldsymbol{K}^Q \right)^\top \right) \right) \boldsymbol{N} \boldsymbol{z}_t \\ &= \boldsymbol{z}_t^\top \boldsymbol{Q} \boldsymbol{z}_t \end{aligned}$$

where

$$\boldsymbol{Q} \equiv \frac{1}{2} \boldsymbol{N}^\top \left(2 \frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{\Sigma} \circ \left(\boldsymbol{K}^Q + \left(\boldsymbol{K}^Q \right)^\top \right) \right) \boldsymbol{N}$$

A.4 Comparative Statics in a Symmetric Economy

This subsection shows how the two ownership matrices shift the operating-profit return that enters a firm's R&D objective in a symmetric economy with $n \geq 2$ firms. Let every off-diagonal entry of $\boldsymbol{\Sigma}$, $\boldsymbol{K}^Q$, and $\boldsymbol{K}^R$ equal σ, k_Q, and k_R, respectively, with $0 < \sigma < 1$ and $k_Q, k_R \in [0, 1]$. By Section A.1, knowledge capital and quantities satisfy $\boldsymbol{z} = \boldsymbol{G}\boldsymbol{q}$ with

$$\boldsymbol{G} = R\boldsymbol{I} + d_2 \boldsymbol{J}, \quad d_1 \equiv 2\frac{\zeta}{L} + 2, \quad d_2 \equiv 2\frac{\zeta}{L} + \sigma \left(1 + k_Q\right), \quad R \equiv d_1 - d_2 = 2 - \sigma \left(1 + k_Q\right) > 0.$$

At a symmetric state $z_i = z > 0$ for all i, quantities are $q_i = z/S$ with $S \equiv d_1 + (n-1)\, d_2$. Because S rises with k_Q, stronger production ownership lowers every firm's quantity, $\partial q_i / \partial k_Q = -(n-1)\, \sigma z / S^2 < 0$, which softens competition. Let

$$\mathcal{M} \equiv \frac{\partial}{\partial z_1} \left(\pi_1 + k_R \sum_{h \neq 1} \pi_h \right)$$

denote the marginal operating-profit return to firm 1's knowledge capital in its R&D objective, evaluated at the symmetric state, holding the other firms' knowledge capital fixed and letting quantities adjust.

Proposition 2. *In the symmetric economy with $n \geq 2$ firms, (i) $\partial \mathcal{M} / \partial k_R < 0$ and (ii) $\partial \mathcal{M} / \partial k_Q \geq 0$, with strict inequality if $k_R < 1$.*

Proof. The matrix $\boldsymbol{G}^{-1} = R^{-1}(\boldsymbol{I} - (d_2/S)\boldsymbol{J})$ gives $\partial q_1/\partial z_1 = (d_1 + (n-2)d_2)/(SR)$ and $\partial q_j/\partial z_1 = -d_2/(SR)$ for $j \neq 1$. By Section A.2, $\pi_i = q_i^2 + u\, q_i \sum_{j\neq i} q_j$ with $u \equiv k_Q\sigma$. Differentiating and evaluating at the symmetric state gives

$$\mathcal{M} = \frac{zB}{S^2R},$$
$$B \equiv 2(d_1 + (n-2)d_2) + (n-1)u(R + (n-2)d_2) - (n-1)k_R[2d_2 + u((n-2)d_2 - R)].$$

For part (i), only B depends on k_R, and

$$\frac{\partial B}{\partial k_R} = -(n-1)\left[4\frac{\zeta}{L} + 2\sigma + 2(n-2)\frac{\zeta}{L}u + (n-1)u(\sigma + u)\right] < 0.$$

For part (ii), because $\partial S/\partial u = n-1$ and $\partial R/\partial u = -1$,

$$\frac{\partial \mathcal{M}}{\partial k_Q} = \sigma\frac{\partial \mathcal{M}}{\partial u} = \frac{\sigma z H}{S^3R^2}, \qquad H \equiv B_u SR + B(S - 2(n-1)R), \qquad B_u \equiv \frac{\partial B}{\partial u}.$$

Since H is linear in k_R, it suffices to sign H at $k_R = 1$ and $k_R = 0$. At $k_R = 1$, $B = 2R[1 + (n-1)u]$ and $B_u = 2[(n-1)R - 1 - (n-1)u]$, so

$$H = 2(n-1)R^2[S - 2 - 2(n-1)u] = 2(n-1)R^2\left[2n\frac{\zeta}{L} + (n-1)\sigma\left(1 - k_Q\right)\right] \geq 0.$$

At $k_R = 0$, substituting the definitions of d_1, d_2, and R and expanding gives

$$H = (n-1)\left[h_1 + (n-1)h_2 + (n-1)^2 h_3\right],$$

where

$$h_1 = 4\frac{\zeta}{L}R^2 + \sigma d_1^2,$$
$$h_2 = \left(2\frac{\zeta}{L} + \sigma\right)R^2 + \left(2\frac{\zeta}{L} + 3u\right)Rd_2 + (\sigma + u)d_2^2,$$
$$h_3 = \left[\left(2\frac{\zeta}{L} + \sigma\right)R + ud_2\right]d_2.$$

Every term is nonnegative and $\sigma d_1^2 > 0$, so $H > 0$. Hence $H > 0$ for $k_R \in [0, 1)$. □

Part (i) states that placing weight on rivals' profits lowers the return to a firm's own knowledge capital, because a firm's knowledge takes profit from its rivals. Part (ii) states that softer competition weakly raises this return. The symmetric economy abstracts from

technology spillovers and holds the other firms' knowledge capital fixed; in the full model, spillovers and the R&D of all firms determine the aggregate effects.

A.5 Derivation of Dynamic Riccati and Lyapunov Equations

This subsection derives the equations used to solve the dynamic R&D game and to evaluate welfare after the equilibrium policy is obtained. Guess that firm i's value function is quadratic,

$$V^i(\boldsymbol{z}) = \boldsymbol{z}^\top \boldsymbol{X}^i \boldsymbol{z},$$

with $\boldsymbol{X}^i$ symmetric. The first-order condition with respect to x_i in equation (21) gives

$$x_i = \mu \left(\boldsymbol{X}_i^i\right)^\top \boldsymbol{z},$$

where $\boldsymbol{X}_i^i$ is the i-th column of $\boldsymbol{X}^i$. Stacking these policy coefficients yields $\boldsymbol{x} = \mu \widetilde{\boldsymbol{X}} \boldsymbol{z}$, where

$$\widetilde{\boldsymbol{X}} \equiv \begin{bmatrix} \boldsymbol{X}_1^1 & \cdots & \boldsymbol{X}_n^n \end{bmatrix}^\top.$$

Thus the drift of knowledge capital under the policy profile is

$$\boldsymbol{\Phi} \boldsymbol{z} = \left(\boldsymbol{\Omega} - \delta \boldsymbol{I} + \mu^2 \widetilde{\boldsymbol{X}}\right) \boldsymbol{z}.$$

Substituting the quadratic guess and the policy profile into firm i's HJB gives

$$\rho\, \boldsymbol{z}^\top \boldsymbol{X}^i \boldsymbol{z} = \boldsymbol{z}^\top \boldsymbol{Q}^i \boldsymbol{z} - \mu^2 \sum_j \kappa_{ij}^R \boldsymbol{z}^\top \boldsymbol{X}_j^j \left(\boldsymbol{X}_j^j\right)^\top \boldsymbol{z} + \boldsymbol{z}^\top \left(\boldsymbol{\Phi}^\top \boldsymbol{X}^i + \boldsymbol{X}^i \boldsymbol{\Phi}\right) \boldsymbol{z}.$$

Since this equality must hold for all $\boldsymbol{z}$, firm i's matrix solves

$$0 = \boldsymbol{Q}^i - \mu^2 \sum_j \kappa_{ij}^R \boldsymbol{X}_j^j \left(\boldsymbol{X}_j^j\right)^\top + \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right)^\top \boldsymbol{X}^i + \boldsymbol{X}^i \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right).$$

These are the stacked algebraic Riccati equations (27).

Once the equilibrium policy $\boldsymbol{x} = \mu \widetilde{\boldsymbol{X}} \boldsymbol{z}$ is known, household welfare evaluates a given policy rather than solving an optimization problem, so its matrix solves a Lyapunov rather than a Riccati equation. With $V(\boldsymbol{z}) = \boldsymbol{z}^\top \boldsymbol{X} \boldsymbol{z}$ and $V_{\boldsymbol{z}}(\boldsymbol{z}) = 2\boldsymbol{z}^\top \boldsymbol{X}$, the household HJB of Section 2.3.1 becomes

$$\rho\, \boldsymbol{z}^\top \boldsymbol{X} \boldsymbol{z} = \boldsymbol{z}^\top \left(\boldsymbol{Q} - \mu^2 \widetilde{\boldsymbol{X}}^\top \widetilde{\boldsymbol{X}}\right) \boldsymbol{z} + 2\boldsymbol{z}^\top \boldsymbol{X} \boldsymbol{\Phi} \boldsymbol{z} = \boldsymbol{z}^\top \left(\boldsymbol{Q} - \mu^2 \widetilde{\boldsymbol{X}}^\top \widetilde{\boldsymbol{X}} + \boldsymbol{X}\boldsymbol{\Phi} + \boldsymbol{\Phi}^\top \boldsymbol{X}\right) \boldsymbol{z},$$

Table A.1: Growth Rate of Variables on the Balanced Growth Path

Growth Rate	Variable
0	$l_{i,t}$
g	$a_{i,t}$, $b_{i,t}$, $z_{i,t}$, $q_{i,t}$, $x_{i,t}$, $p_{i,t}/P_t$
$2g$	$\pi_{i,t}$, C_t, Y_t, w_t/P_t

Note: Growth rates are evaluated along the balanced-growth-path equilibrium.

where the second equality symmetrizes the quadratic form. Because the matrices on both sides are symmetric and the equality holds for all $\boldsymbol{z}$, $\boldsymbol{X}$ solves the Lyapunov equation (29).

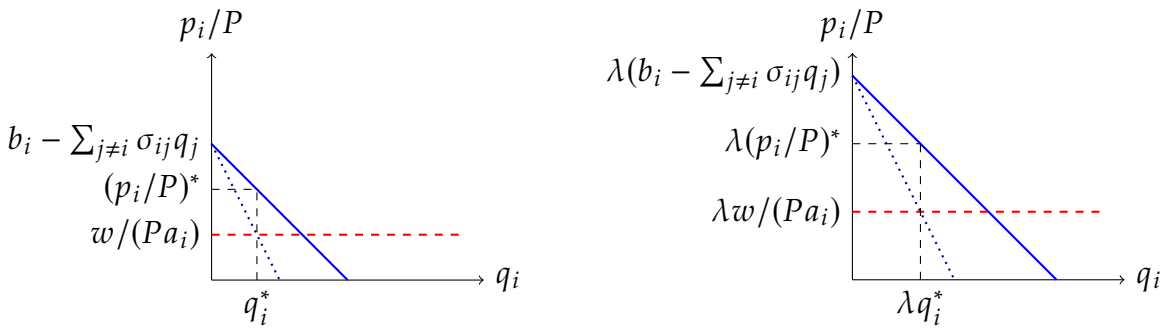
A.6 Balanced Growth Path

Proof of Proposition 1. Choose $c > 0$ large enough that $\boldsymbol{A} \equiv \boldsymbol{\Phi} + c\boldsymbol{I}$ is nonnegative. Adding a multiple of the identity does not change the off-diagonal graph, so $\boldsymbol{A}$ is irreducible under Assumption 2. By the Perron–Frobenius theorem, $\boldsymbol{A}$ has a simple real eigenvalue $r(\boldsymbol{A})$ equal to its spectral radius, with a strictly positive eigenvector $\bar{\boldsymbol{z}}$. Hence $g = r(\boldsymbol{A}) - c$ is a real eigenvalue of $\boldsymbol{\Phi}$ with the same eigenvector, and every other eigenvalue of $\boldsymbol{\Phi}$ has a smaller real part. Because $\exp(\boldsymbol{\Phi}t) = \exp(-ct)\exp(\boldsymbol{A}t)$ and $\exp(\boldsymbol{A}t)$ is strictly positive for $t > 0$, the direction of $\exp(\boldsymbol{\Phi}t)\,\boldsymbol{z}_0$ converges to that of $\bar{\boldsymbol{z}}$ for any strictly positive $\boldsymbol{z}_0$, and the knowledge capital of all firms grows asymptotically at rate g. □

Figure A.1 illustrates a partial equilibrium to provide an intuitive understanding of how a balanced-growth-path equilibrium is achieved. In equilibrium, as quality (b_i), quantity (q_i), real price (p_i/P), and real marginal cost ($w/(Pa_i)$) all grow at the same rate g, the partial-equilibrium diagram expands homothetically at this common rate along both axes. Consequently, ratios such as revenue to cost and consumer surplus to producer surplus remain constant along the path. The figure also suggests that growth in both labor productivity a_i and quality b_i is necessary. For example, if only quality b_i grows and labor productivity a_i does not, it is impossible for $q_i = a_i l_i$ to exhibit sustained growth because labor supply is inelastic.

Moreover, Figure A.1 indicates why demand elasticity and the elasticity of substitution across goods remain constant along the balanced growth path. Because we assume a linear inverse demand system rather than CES demand, these elasticities would generally vary as the equilibrium point moves along the demand curve. Along the balanced growth path, however, the diagram scales proportionally, so the equilibrium point keeps its relative position on the demand curve and both elasticities remain constant.

Figure A.1: Partial Equilibrium Diagram and Its Proportional Growth



Note: The left panel shows one firm's static equilibrium, with its residual demand as a solid line, marginal revenue as a dotted line, and real marginal cost as a thick dashed line; thin dashed lines mark the equilibrium price and quantity. The right panel shows that over an interval Δt the equilibrium real price, quantity, real marginal cost, and demand intercept all scale by the growth factor $\lambda = e^{g\Delta t}$.

A.7 Social Planner

Unlike the R&D-only planner of Section 2.4.2, the social planner chooses both production and R&D to maximize the representative household's lifetime utility, subject to technological constraints. Neither ownership matrix enters this problem, so the social optimum is the same for every ownership structure. As in the market equilibrium, the planner's problem decomposes into a static and a dynamic component.

Static Problem Following Hopenhayn and Okumura (2026), given knowledge capital $\boldsymbol{z}_t$, the planner maximizes final-good production at each instant,

$$\max_{\boldsymbol{a}_t, \boldsymbol{b}_t, \boldsymbol{l}_t, \boldsymbol{q}_t} \; Y_t = \boldsymbol{q}_t^\top \boldsymbol{b}_t - \frac{1}{2} \boldsymbol{q}_t^\top \boldsymbol{\Sigma} \boldsymbol{q}_t,$$

subject to

$$\begin{aligned} q_{i,t} &= a_{i,t} l_{i,t} \quad i \in \{1, 2, \ldots, n\}, \\ z_{i,t} &= \zeta a_{i,t} + b_{i,t} \quad i \in \{1, 2, \ldots, n\}, \\ L &= \sum_i l_{i,t}. \end{aligned} \tag{36}$$

The first constraint is each firm's production technology; the second allocates each firm's knowledge capital between labor productivity and product quality; the third clears the

labor market. The Lagrangian is

$$\mathcal{L} = \boldsymbol{q}_t^\top (\boldsymbol{z}_t - \zeta \boldsymbol{a}_t) - \frac{1}{2} \boldsymbol{q}_t^\top \boldsymbol{\Sigma} \boldsymbol{q}_t - \nu_t \left\{ \sum_i \frac{q_{i,t}}{a_{i,t}} - L \right\}$$

The first-order condition with respect to $a_{i,t}$ gives

$$a_{i,t} = \sqrt{\frac{\nu_t}{\zeta}} \tag{37}$$

and the knowledge constraint gives

$$b_{i,t} = z_{i,t} - \sqrt{\zeta \nu_t} \tag{38}$$

Inserting equation (37) into equation (36), we obtain

$$\sqrt{\zeta \nu_t} = \frac{\zeta}{L} \sum_i q_{i,t} \tag{39}$$

In vector form,

$$\sqrt{\zeta \nu_t} \mathbf{1} = \frac{\zeta}{L} \left(\sum_i q_{i,t} \right) \mathbf{1} = \frac{\zeta}{L} \boldsymbol{J} \boldsymbol{q}_t. \tag{40}$$

The first-order condition with respect to $\boldsymbol{q}_t$ gives

$$\boldsymbol{z}_t - \zeta \boldsymbol{a}_t - \boldsymbol{\Sigma} \boldsymbol{q}_t = \nu_t \left(1/a_{1,t}, \ldots, 1/a_{n,t}\right)^\top \tag{41}$$

Inserting equation (37) and then equation (40), we obtain $\boldsymbol{z}_t = 2\sqrt{\zeta \nu_t} \mathbf{1} + \boldsymbol{\Sigma} \boldsymbol{q}_t = \left(2\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} \right) \boldsymbol{q}_t$. The planner's quantities are therefore linear in knowledge capital,

$$\boldsymbol{q}_t^* = \boldsymbol{N}^* \boldsymbol{z}_t, \qquad \boldsymbol{N}^* \equiv \left(2\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} \right)^{-1}. \tag{42}$$

Inserting equations (38) and (40) into equation (4) and using equation (42), output is a quadratic form in knowledge capital,

$$Y_t^* = \boldsymbol{q}_t^{*\top} \left(\left(2\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} \right) \boldsymbol{q}_t^* - \frac{\zeta}{L} \boldsymbol{J} \boldsymbol{q}_t^* \right) - \frac{1}{2} \boldsymbol{q}_t^{*\top} \boldsymbol{\Sigma} \boldsymbol{q}_t^* = \boldsymbol{q}_t^{*\top} \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} \right) \boldsymbol{q}_t^* = \frac{1}{2} \boldsymbol{z}_t^\top \boldsymbol{N}^* \boldsymbol{z}_t.$$

Dynamic Problem The social planner solves

$$V^*(z_0) \equiv \max_{\{x_t\}_{t\geq 0}} \int_0^\infty \exp(-\rho t) \left\{ \frac{1}{2} z_t^\top N^* z_t - x_t^\top x_t \right\} dt,$$

subject to the law of motion of knowledge capital (8). The HJB equation is

$$\rho V^*(z) = \max_x \left\{ \frac{1}{2} z^\top N^* z - x^\top x + V_z^*(z) \left[(\Omega - \delta I) z + \mu x \right] \right\} \tag{43}$$

The first two terms on the right-hand side are output net of total R&D cost, the household's instantaneous consumption, and the third term is the change in the household's value from the accumulation of knowledge capital.

Because the social planner's problem is also linear-quadratic, the value is quadratic in knowledge capital,

$$V^*(z) = z^\top X^* z \tag{44}$$

The planner's R&D choice is not restricted in sign, so the socially optimal allocation of R&D is a linear function of knowledge capital z, as in the market equilibrium,

$$x = \mu X^* z \tag{45}$$

The law of motion for knowledge capital z_t becomes a linear system with constant coefficients,

$$\dot{z}_t = \Phi^* z_t$$

where the optimal technology transition Φ^* is given by

$$\Phi^* \equiv (\Omega - \delta I) + \mu^2 X^*$$

Inserting equation (44) and its derivative with respect to z and equation (45) into equation (43), we obtain a single algebraic Riccati equation

$$0 = \frac{1}{2} N^* - \mu^2 (X^*)^2 + X^* \left(\Phi^* - \frac{\rho}{2} I \right) + \left(\Phi^* - \frac{\rho}{2} I \right)^\top X^* \tag{46}$$

When equation (46) has a stabilizing solution, one for which all eigenvalues of $\Phi^* - \frac{\rho}{2} I$ have strictly negative real parts, social welfare under the optimal allocation is $V^*(z_0) = z_0^\top X^* z_0$.

A.8 Stochastic Extension

This subsection adds idiosyncratic shocks to knowledge capital. Let $\boldsymbol{W}_t$ be a vector of n independent standard Brownian motions, and let knowledge capital follow

$$d\boldsymbol{z}_t = \left((\boldsymbol{\Omega} - \delta \boldsymbol{I})\, \boldsymbol{z}_t + \mu \boldsymbol{x}_t\right) dt + \gamma\, \mathrm{diag}\left(\boldsymbol{z}_t\right) d\boldsymbol{W}_t,$$

where $\gamma \geq 0$ scales the shocks and $\mathrm{diag}\left(\boldsymbol{z}_t\right)$ is the diagonal matrix with $\boldsymbol{z}_t$ on its diagonal. Firms, the household, and the planners maximize expected discounted payoffs. The static problem is unchanged because it depends only on current knowledge capital.

For a quadratic value function $V\left(\boldsymbol{z}\right) = \boldsymbol{z}^\top \boldsymbol{X} \boldsymbol{z}$ with $\boldsymbol{X}$ symmetric, Itô's lemma adds

$$\frac{1}{2}\mathrm{Tr}\left[\gamma^2 \mathrm{diag}\left(\boldsymbol{z}\right)\left(2\boldsymbol{X}\right)\mathrm{diag}\left(\boldsymbol{z}\right)\right] = \gamma^2 \sum_k X_{kk} z_k^2 = \gamma^2 \boldsymbol{z}^\top \mathcal{D}\left(\boldsymbol{X}\right)\boldsymbol{z}$$

to the right-hand side of the corresponding HJB equation, where $\mathcal{D}\left(\boldsymbol{X}\right)$ is the diagonal matrix that retains the diagonal of $\boldsymbol{X}$. Only the diagonal enters because each shock affects a single firm's knowledge capital. The term is quadratic in $\boldsymbol{z}$, so value functions remain quadratic and R&D policies remain linear, $\boldsymbol{x} = \mu \widetilde{\boldsymbol{X}} \boldsymbol{z}$. The stacked Riccati equations (27) become

$$0 = \boldsymbol{Q}^i - \mu^2 \sum_j \kappa_{ij}^R \boldsymbol{X}_j^j \left(\boldsymbol{X}_j^j\right)^\top + \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right)^\top \boldsymbol{X}^i + \boldsymbol{X}^i \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right) + \gamma^2 \mathcal{D}\left(\boldsymbol{X}^i\right).$$

Similarly, the term $\gamma^2 \mathcal{D}\left(\boldsymbol{X}\right)$ is added to the Lyapunov equation (29), and $\gamma^2 \mathcal{D}\left(\boldsymbol{X}^{**}\right)$ and $\gamma^2 \mathcal{D}\left(\boldsymbol{X}^{*}\right)$ are added to the planners' Riccati equations (31) and (46).

Under the linear policies, the second moments $\boldsymbol{M}_t \equiv \mathbb{E}\left[\boldsymbol{z}_t \boldsymbol{z}_t^\top\right]$ evolve according to

$$\dot{\boldsymbol{M}}_t = \boldsymbol{\Phi} \boldsymbol{M}_t + \boldsymbol{M}_t \boldsymbol{\Phi}^\top + \gamma^2 \mathcal{D}\left(\boldsymbol{M}_t\right).$$

Discounted payoffs are finite if the linear map $\boldsymbol{M} \mapsto \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right)\boldsymbol{M} + \boldsymbol{M}\left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right)^\top + \gamma^2 \mathcal{D}\left(\boldsymbol{M}\right)$ has eigenvalues with strictly negative real parts, which reduces to the stability condition in the main text when $\gamma = 0$. Expected knowledge capital follows $\frac{d}{dt}\mathbb{E}\left[\boldsymbol{z}_t\right] = \boldsymbol{\Phi}\,\mathbb{E}\left[\boldsymbol{z}_t\right]$, so Proposition 1 characterizes the balanced growth of expected knowledge capital, with $\boldsymbol{\Phi}$ obtained from the Riccati equations above.

A.9 Derivation of Stochastic Process of Output

Under the stochastic extension, knowledge capital follows $d\boldsymbol{z}_t = \boldsymbol{\Phi}\boldsymbol{z}_t dt + \gamma\,\mathrm{diag}\,(\boldsymbol{z}_t)\,d\boldsymbol{W}_t$ and output is $Y_t = \boldsymbol{z}_t^\top \boldsymbol{Q}\boldsymbol{z}_t$. Because the gradient of Y_t with respect to $\boldsymbol{z}_t$ is $2\boldsymbol{Q}\boldsymbol{z}_t$ and its Hessian is $2\boldsymbol{Q}$, Itô's lemma gives

$$\begin{aligned} dY_t &= \left\{2\boldsymbol{z}_t^\top \boldsymbol{Q}\boldsymbol{\Phi}\boldsymbol{z}_t + \frac{1}{2}\mathrm{Tr}\left[\gamma^2 \mathrm{diag}\,(\boldsymbol{z}_t)\,(2\boldsymbol{Q})\,\mathrm{diag}\,(\boldsymbol{z}_t)\right]\right\} dt + 2\gamma \boldsymbol{z}_t^\top \boldsymbol{Q}\mathrm{diag}\,(\boldsymbol{z}_t)\,d\boldsymbol{W}_t \\ &= \left\{\boldsymbol{z}_t^\top \left(\boldsymbol{Q}\boldsymbol{\Phi} + \boldsymbol{\Phi}^\top \boldsymbol{Q}\right)\boldsymbol{z}_t + \gamma^2 \sum_i z_{i,t}^2 \left[\boldsymbol{Q}\right]_{ii}\right\} dt + 2\gamma \boldsymbol{z}_t^\top \boldsymbol{Q}\mathrm{diag}\,(\boldsymbol{z}_t)\,d\boldsymbol{W}_t. \end{aligned}$$

Applying Itô's lemma to $\log Y_t$ then gives

$$\begin{aligned} d\log Y_t = &\left[\frac{\boldsymbol{z}_t^\top \left(\boldsymbol{Q}\boldsymbol{\Phi} + \boldsymbol{\Phi}^\top \boldsymbol{Q}\right)\boldsymbol{z}_t}{\boldsymbol{z}_t^\top \boldsymbol{Q}\boldsymbol{z}_t} + \gamma^2 \left\{\frac{\sum_i z_{i,t}^2 \left[\boldsymbol{Q}\right]_{ii}}{\boldsymbol{z}_t^\top \boldsymbol{Q}\boldsymbol{z}_t} - 2\frac{\boldsymbol{z}_t^\top \boldsymbol{Q}\mathrm{diag}\,(\boldsymbol{z}_t)^2\,\boldsymbol{Q}\boldsymbol{z}_t}{\left(\boldsymbol{z}_t^\top \boldsymbol{Q}\boldsymbol{z}_t\right)^2}\right\}\right] dt \\ &+ 2\gamma \frac{\boldsymbol{z}_t^\top \boldsymbol{Q}\mathrm{diag}\,(\boldsymbol{z}_t)}{\boldsymbol{z}_t^\top \boldsymbol{Q}\boldsymbol{z}_t} d\boldsymbol{W}_t \end{aligned}$$

Appendix B Computation and Additional Results

B.1 Solution Algorithm

To solve the equilibrium, we solve the system of Riccati equations backward. Specifically, given $\begin{bmatrix} \boldsymbol{X}_\tau^1 & \cdots & \boldsymbol{X}_\tau^n \end{bmatrix}$, update $\begin{bmatrix} \boldsymbol{X}_{\tau-\Delta}^1 & \cdots & \boldsymbol{X}_{\tau-\Delta}^n \end{bmatrix}$ by

$$-\frac{\boldsymbol{X}_\tau^i - \boldsymbol{X}_{\tau-\Delta}^i}{\Delta} = \boldsymbol{Q}^i - \mu^2 \sum_j \kappa_{ij}^R \boldsymbol{X}_{j,\tau}^j \left(\boldsymbol{X}_{j,\tau}^j\right)^\top + \left(\boldsymbol{\Phi}_\tau - \frac{\rho}{2}\boldsymbol{I}\right)^\top \boldsymbol{X}_\tau^i + \boldsymbol{X}_\tau^i \left(\boldsymbol{\Phi}_\tau - \frac{\rho}{2}\boldsymbol{I}\right)$$

where $\boldsymbol{\Phi}_\tau = \boldsymbol{\Omega} + \mu^2 \widetilde{\boldsymbol{X}}_\tau - \delta \boldsymbol{I}$ and $\boldsymbol{X}_{j,\tau}^j$ is the jth column of $\boldsymbol{X}_\tau^j$. We iterate the Riccati equations backward from zero terminal values, $\boldsymbol{X}^i = \boldsymbol{0}$, which selects the stable limit of the finite-horizon equilibria (Cartigny and Michel, 2003). The iteration stores the n matrices $\boldsymbol{X}^i$, so its memory grows as $O\left(n^3\right)$, and each update requires $O\left(n^4\right)$ operations. With 638 firms, the equilibrium solves in 19 seconds on an Apple M4 Pro laptop with 24 GB of memory, using six threads. Table A.2 compares this scale with a dynamic oligopoly that discretizes each firm's productivity.

Table A.2: Computational Complexity and Scale

Model	Storage	Arithmetic per update	Number of firms	Productivity state space
Cavenaile et al. (2026)	$O(k^n)$	–	4	Discrete grid
Our model	$O(n^3)$	$O(n^4)$	638	Continuous

Note: k is the number of grid points for each firm's productivity, so a discretized state space has k^n points. For our model, storage counts the coefficients of the n matrices $\boldsymbol{X}^i$.

B.2 Interior Linear Solution and Nonnegativity

The allocations derived in Sections A.1 and A.5 characterize the interior linear solution. They assume that the implied quantities and R&D efforts are nonnegative and that the matrix $2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{K}^Q \circ \boldsymbol{\Sigma}$, whose inverse is $\boldsymbol{N}$, is nonsingular. The quantitative analysis uses the interior linear solution, which preserves the tractability of the Riccati and Lyapunov systems. We check nonnegativity at the observed state in the four economies of Section 4.5 and under the R&D-only planner. Among these economies, negative quantities arise only when production ownership is at its 2004 intensity, $\boldsymbol{K}^Q = \boldsymbol{K}_{2004}$, where 26 firms have negative quantities, accounting for 0.19 percent of absolute quantity. They also arise in the other economies whose production ownership differs from $\boldsymbol{K}$, for 51 firms under fully dispersed ownership and for at most 32 firms under the governance models of Section 5. Negative R&D arises for at most 40 firms in any of the four economies (8 in the observed economy) and for 81 firms under the R&D-only planner. To gauge the consequences, we hold the R&D rules fixed, re-solve production with quantities constrained to be nonnegative, and replace each firm's R&D effort x_i by $\max\{0, x_i\}$ in both the R&D cost and the accumulation of knowledge capital. This changes discounted consumption by at most 0.013 percent and leaves the signs of the effects on growth and welfare in Tables 3 and 5 unchanged.

B.3 Business-Stealing and Spillover Channels

This subsection defines the calculation behind the comparison of the two forces in Section 4.4. It holds R&D policies fixed at the observed state, sets production ownership to $\boldsymbol{K}^Q = \boldsymbol{K}$, and approximates to first order how raising R&D ownership from $\boldsymbol{K}_{2004}$ to $\boldsymbol{K}$ changes the value that owners attach to a unit of each firm's knowledge.

By Section A.2, firm h's gross profit is $\pi_h = q_h \sum_j \kappa^Q_{hj}\sigma_{hj}q_j$ with $\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$. Let $\boldsymbol{A} \equiv \boldsymbol{K}^Q \circ \boldsymbol{\Sigma}$ and let $\boldsymbol{D}$ collect the derivatives of static profits with respect to knowledge capital,

$D_{hm} \equiv \partial \pi_h / \partial z_m$. Differentiating gives

$$\boldsymbol{D} = \operatorname{diag}\left(\boldsymbol{A}\boldsymbol{q}\right) \boldsymbol{N} + \operatorname{diag}\left(\boldsymbol{q}\right) \boldsymbol{A}\boldsymbol{N}.$$

The owners of firm i weight firm h's profit by κ^R_{ih}, so a unit of firm m's knowledge raises the flow objective of firm i by $G_{im} \equiv \sum_h \kappa^R_{ih} D_{hm}$. Raising R&D ownership by $\Delta \boldsymbol{K} \equiv \boldsymbol{K} - \boldsymbol{K}_{2004}$, with entries $\Delta\kappa_{ih}$, changes these values by $\Delta \boldsymbol{G} = \Delta \boldsymbol{K} \boldsymbol{D}$. Because both matrices have unit diagonals, $\Delta\kappa_{ii} = 0$.

The value of a unit of firm i's own knowledge changes through business stealing by

$$\Delta G_{ii} = \sum_{h \neq i} \Delta\kappa_{ih} D_{hi},$$

the change in the weight on the profit that the knowledge takes from rivals. The derivatives D_{hi} for $h \neq i$ include both product-market rivalry and the interaction through the common real wage, the two links that the technology-spillover economy in Section 4.4 removes. The same unit adds knowledge to firm j at rate $\beta\tilde{\omega}_{ji}$ by equation (8). Counting only these direct spillovers, and with both the unit and the knowledge it spills over depreciating at rate δ and discounted at rate ρ, the unit's own profit flow is worth $G_{ii}/(\rho + \delta)$ to the owners of firm i, and the knowledge it spills over to firm j is worth $\beta\tilde{\omega}_{ji} G_{ij}/(\rho + \delta)^2$. Measured in the same flow units as ΔG_{ii}, the value of the unit therefore changes through spillovers by

$$S_i = \frac{\beta}{\rho + \delta} \sum_{j \neq i} \tilde{\omega}_{ji} \, \Delta G_{ij}.$$

$\Delta G_{ij} = \sum_h \Delta\kappa_{ih} D_{hj}$ includes the change in the weight on the profit that the spilled knowledge takes from firms other than i and j. The source firm i contributes nothing because $\Delta\kappa_{ii} = 0$. Without the other firms, ΔG_{ij} would be $\Delta\kappa_{ij} D_{jj}$, which gives the value S^0_i.

Table A.3 reports $\sum_i w_i S^0_i$ and $\sum_i w_i S_i$ as percents of $-\sum_i w_i \Delta G_{ii}$, with firms weighted by knowledge capital, equally, or by R&D expenditure in the data. The main text uses knowledge-capital weights. Under all three weightings, the rise through spillovers is a small fraction of the fall through business stealing, at most 39 percent if the spilled knowledge took no business and 9 percent once it does, so the comparison in Section 4.4 does not depend on how firms are weighted.

Table A.3: Business-Stealing and Spillover Channels

	Firms weighted by		
	Knowledge capital	Equally	R&D expenditure
Spilled knowledge takes no business	25	23	39
Spilled knowledge takes business	5	4	9

Note: Each entry is the rise through spillovers in the value that owners attach to a unit of a firm's knowledge, as a percent of the fall through business stealing, when R&D ownership rises from $\boldsymbol{K}_{2004}$ to $\boldsymbol{K}$ with production ownership at $\boldsymbol{K}^{Q} = \boldsymbol{K}$. The first row ignores the business that the spilled knowledge takes from firms other than its source and recipient.

Table A.4: Ownership Placement and R&D by Firm Size

Firm size (knowledge capital)	Share of R&D (%)	R&D change, $\boldsymbol{K}$ (%)	R&D change, $\bar{\boldsymbol{K}}$ (%)	Rise in weight, $\boldsymbol{K}$	Rise in weight, $\bar{\boldsymbol{K}}$
Smallest 20%	1	−53.8	−66.8	0.15	0.28
Second 20%	3	−45.4	−54.7	0.18	0.26
Middle 20%	5	−39.8	−42.9	0.21	0.23
Fourth 20%	9	−38.4	−26.4	0.25	0.19
Largest 20%	82	−16.9	−4.6	0.31	0.14

Note: Groups are quintiles of knowledge capital in 2019. Production ownership is $\boldsymbol{K}^{Q} = \boldsymbol{K}$. The share of R&D is each group's share of R&D expenditure with R&D ownership at $\boldsymbol{K}_{2004}$. R&D changes are percent changes in each group's total R&D expenditure from raising R&D ownership from $\boldsymbol{K}_{2004}$ to $\boldsymbol{K}$ or to $\bar{\boldsymbol{K}}$. The rise in weight is the mean, over other firms j, of the increase in κ^{R}_{ij} from its level in $\boldsymbol{K}_{2004}$. Because $\boldsymbol{K}_{2004}$ scales the off-diagonal weights of $\boldsymbol{K}$ by a common factor, the rise under $\boldsymbol{K}$ is proportional to the weight itself.

B.4 Ownership Placement by Firm Size

Table A.4 shows, by firm size, the placement of ownership behind Section 4.4. The observed matrix $\boldsymbol{K}$ raises the weights of the largest firms the most, whereas the uniform matrix $\bar{\boldsymbol{K}}$ raises those of the smallest firms the most. Within every group, R&D falls more in the counterfactual in which the group's weights rise more.